

Controllable Language Generation at Every Scale

Balancing Precision and Resources

Presented by Jillian Fisher
General Exam 12/4/2024

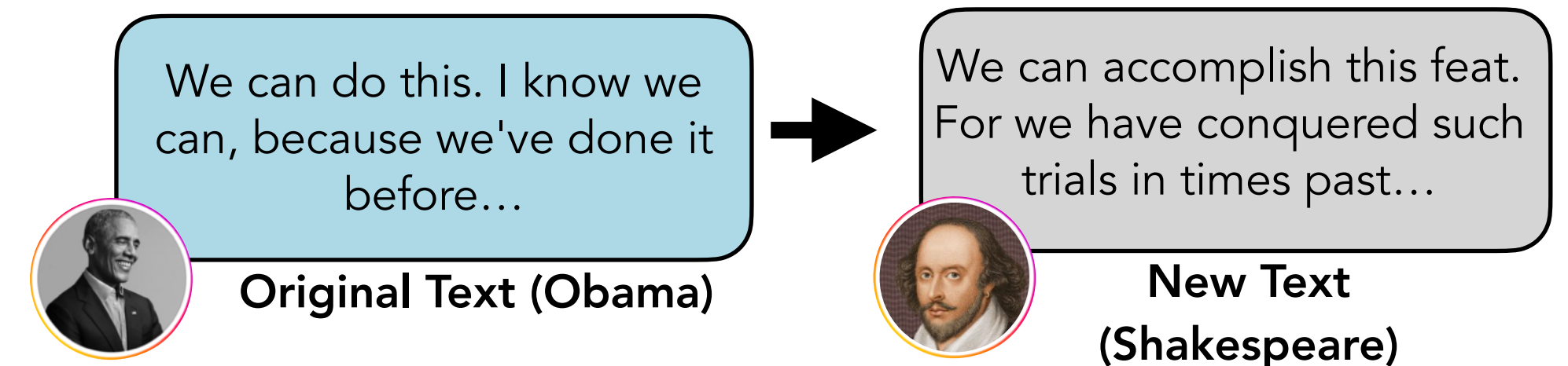


Controllable Generation

Method which directs a model's output to meet specific criteria.

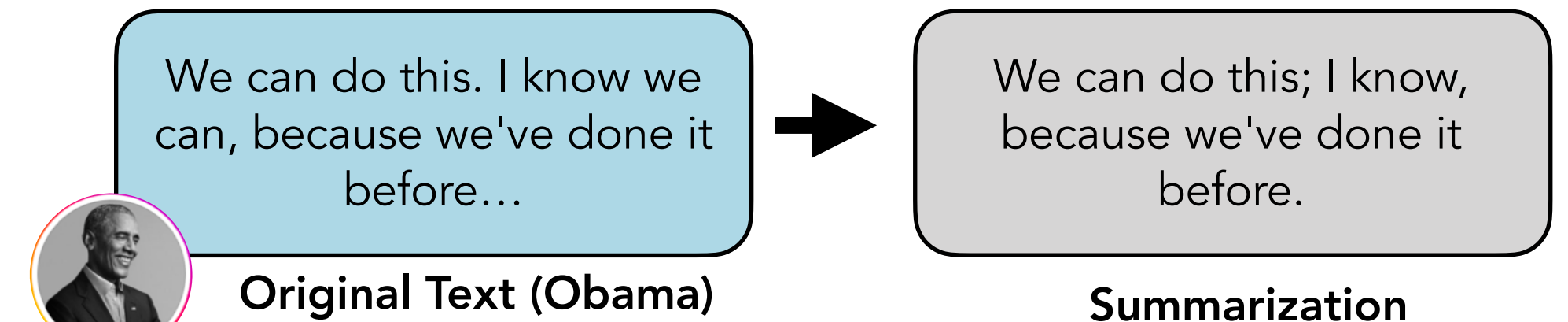
Style Transfer

Criteria: *Target Style*



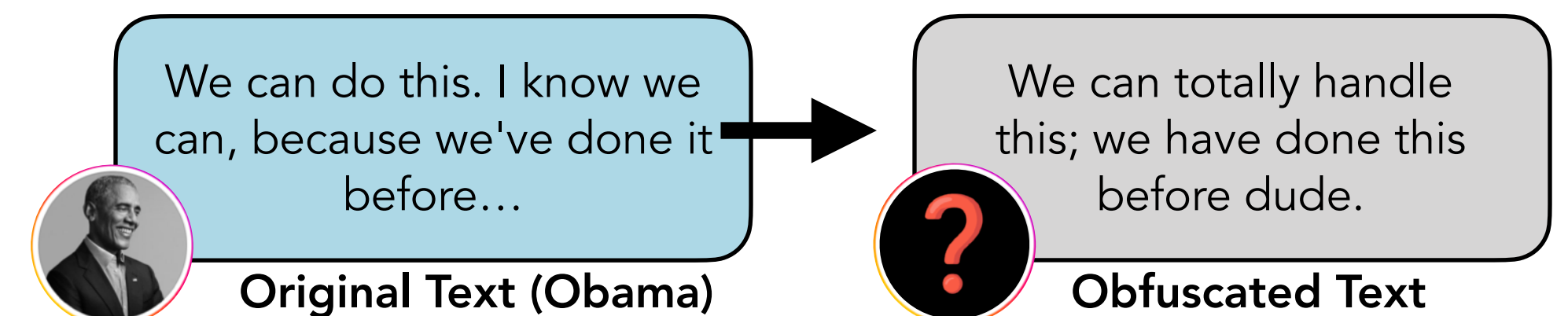
Summarization

Criteria: *Shorter*



Authorship Obfuscation

Criteria: *Not Original Author Style*



Controllable Generation

Authorship
Obfuscation



We can do this. I know we can, because
we've done it before...

Original Text (Obama)



We can totally handle this; we
have done this before dude.

Obfuscated Text

What?

Rewriting text to obscure the original author's identity

Should maintain the content and sentiment

Why?

Blind Review for
Scientific Papers



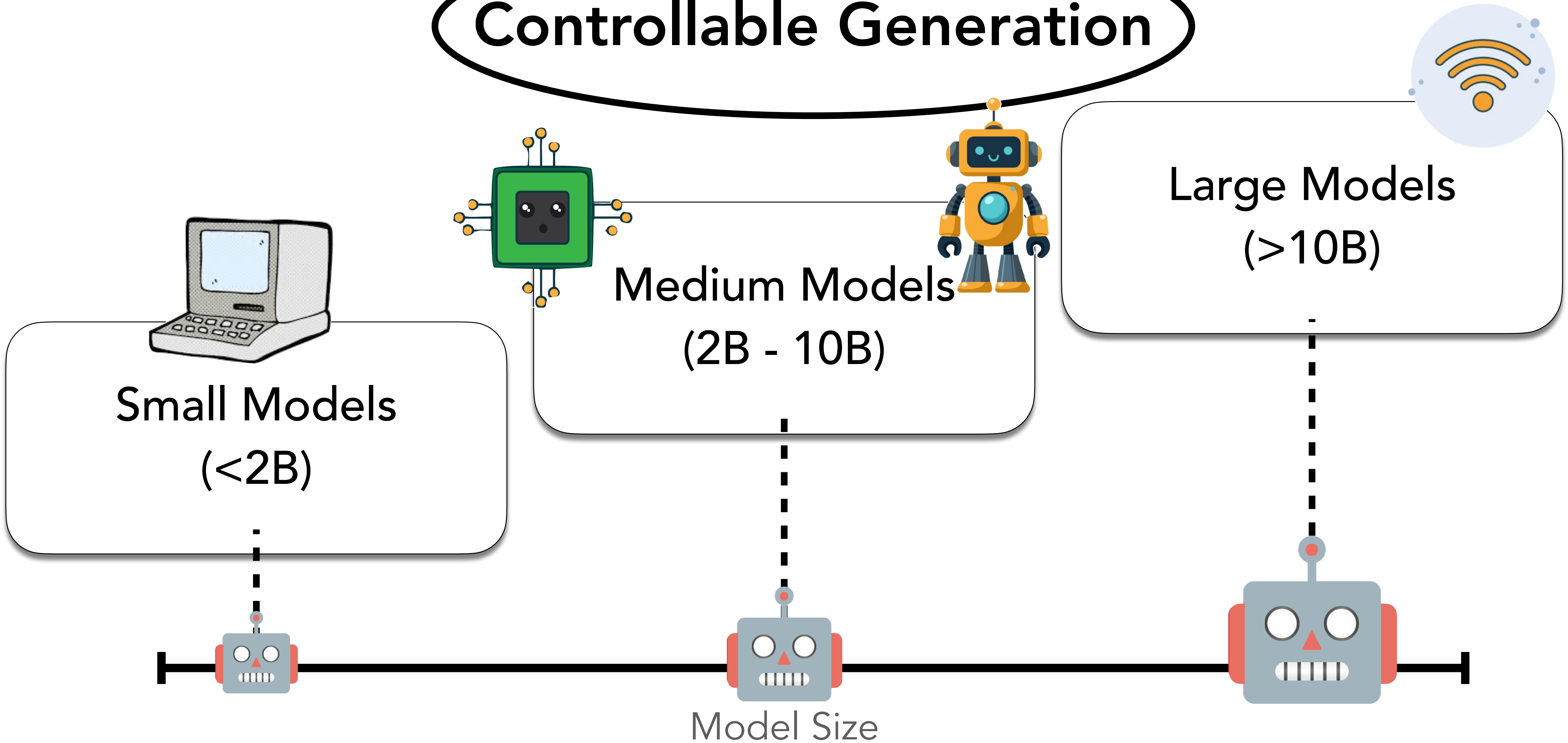
Interaction on Mental
Health Forums



Anonymous
Online Review



Controllable Generation

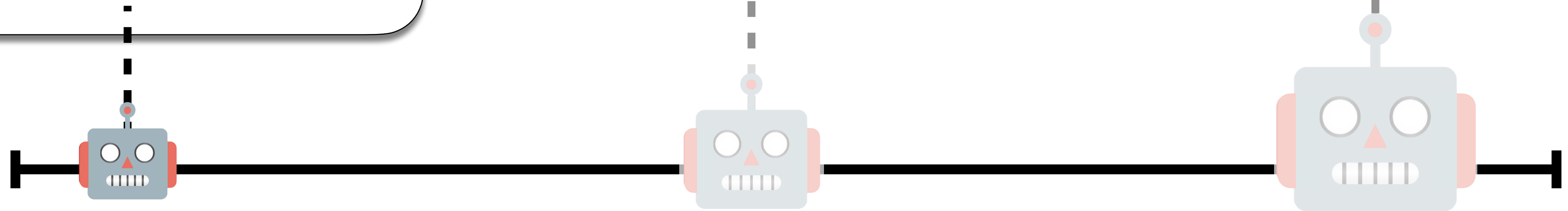


Controllable Generation

Small Models
($<2B$)

Medium Models
($2B - 10B$)

Large Models
($>10B$)



Model Size

Small Models ($<2B$)

Can a small model
method beat larger
scale models?

Pros

Access to internal
data (e.g. logits)

Cheap (training and
inference)

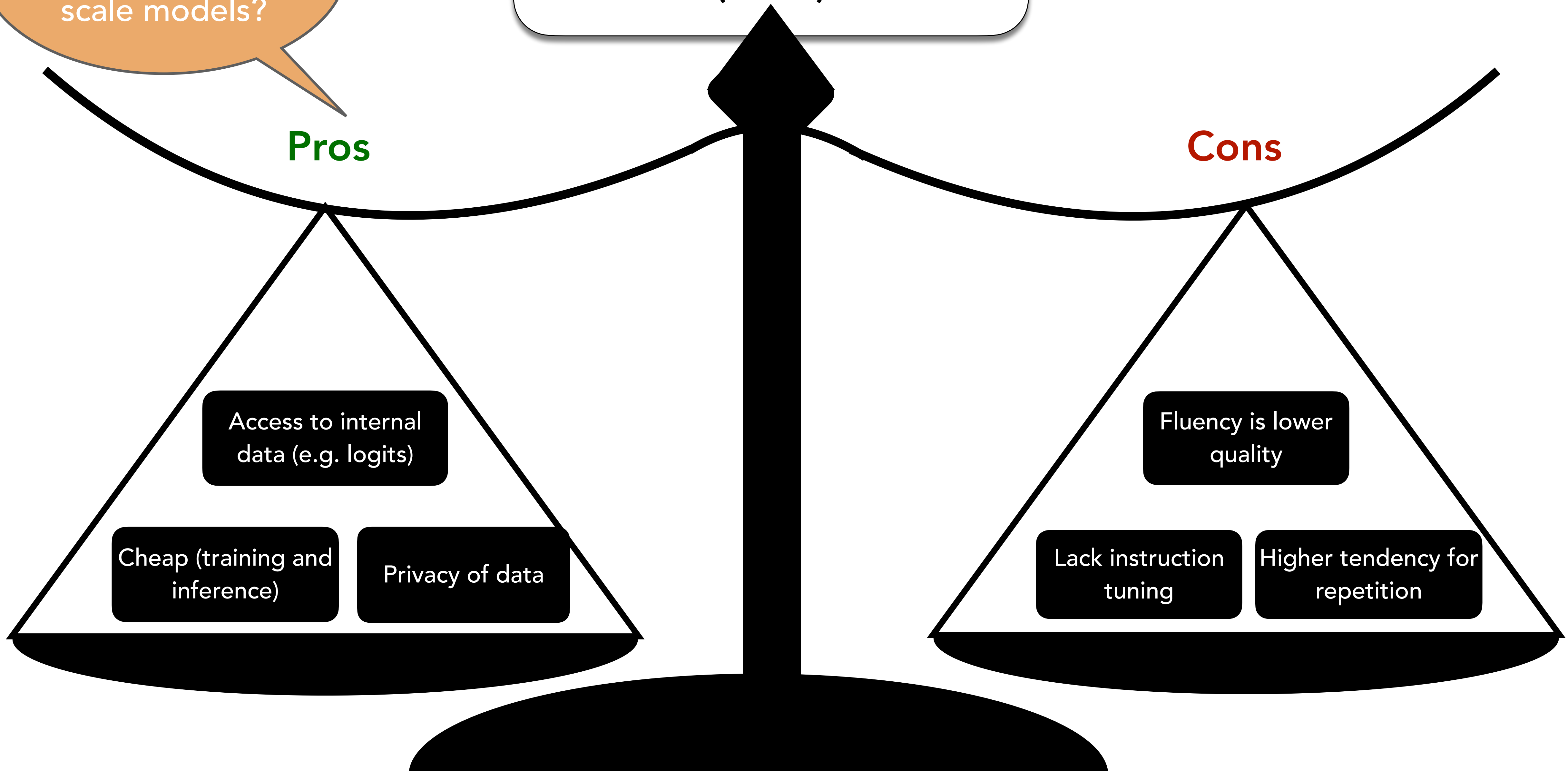
Privacy of data

Cons

Fluency is lower
quality

Lack instruction
tuning

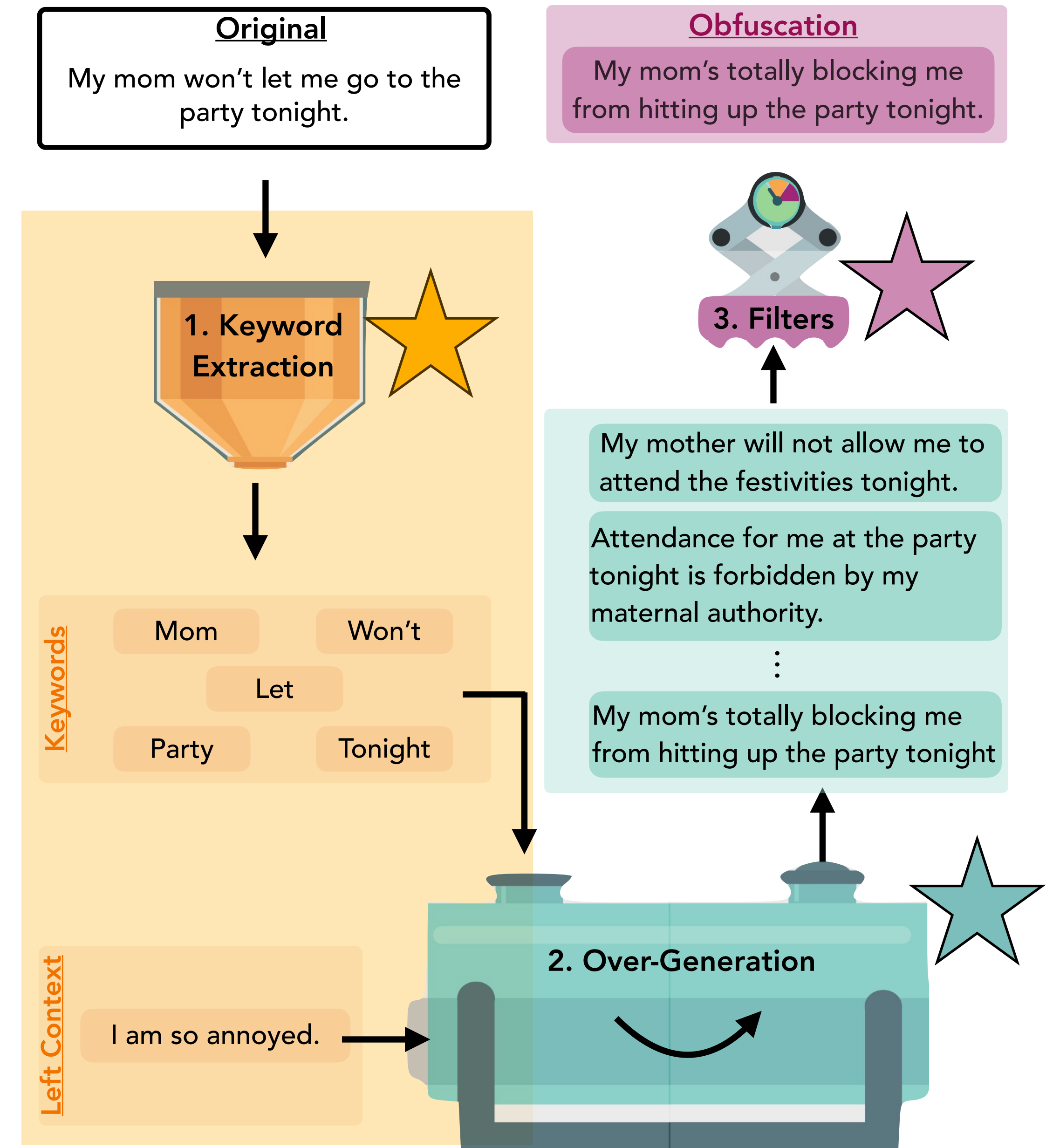
Higher tendency for
repetition



JAMDEC Decoding

Contributions:

- New authorship obfuscation method aimed for small models (JAMDEC Decoding)
- Highlight a new keyword extraction method
- Explore new combined beam-search method
- User-controlled, inference-time algorithm for authorship obfuscation that can be applied to any text and authorship without a separate authorship corpus
- **3 Stage Approach:**
 1. *Keyword Extraction*: Maintain original content
 2. *Over-generation*: Generate diverse outputs that include the keywords
 3. *Filters*: Maintain fluency and content preservation, + user-specified control

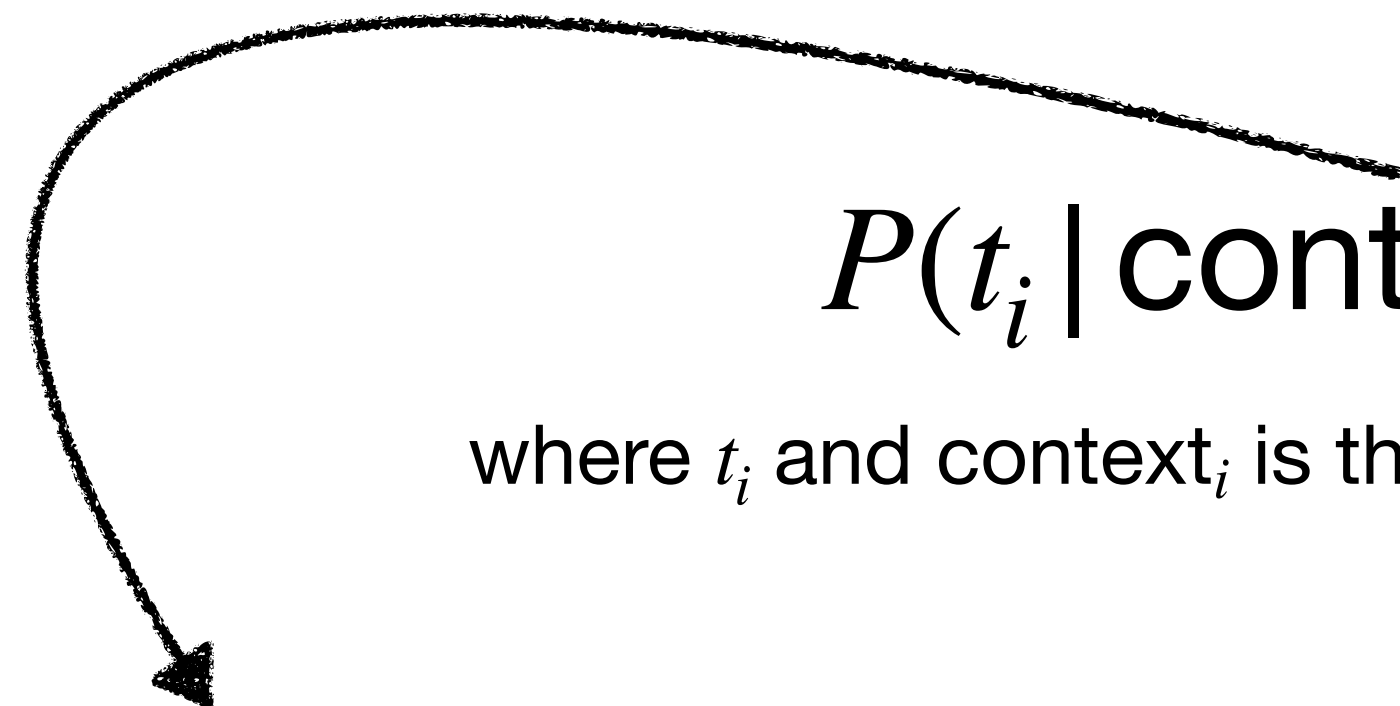


Innovations: Keyword Extraction

- Current methods rely on cosine similarity of a word-embeddings to a document-embedding

New Likelihood-based Method

- Keywords = top-k tokens with the lowest conditional probabilities, as measured by a specific language model



$$P(t_i | \text{context}_i)$$

where t_i and context_i is the token and given context at time i .

Auto-Regressive
(GPT2)

$$P(t_i | t_1, t_2, \dots, t_{i-1})$$

Text-to-Text
(T5)

$$P(t_i | t_1, \dots, t_{i-1}, [MASK], t_{i+1}, \dots, t_n)$$

Original
My mom won't let me go to
the party tonight.

**1. Keyword
Extraction**

Keywords

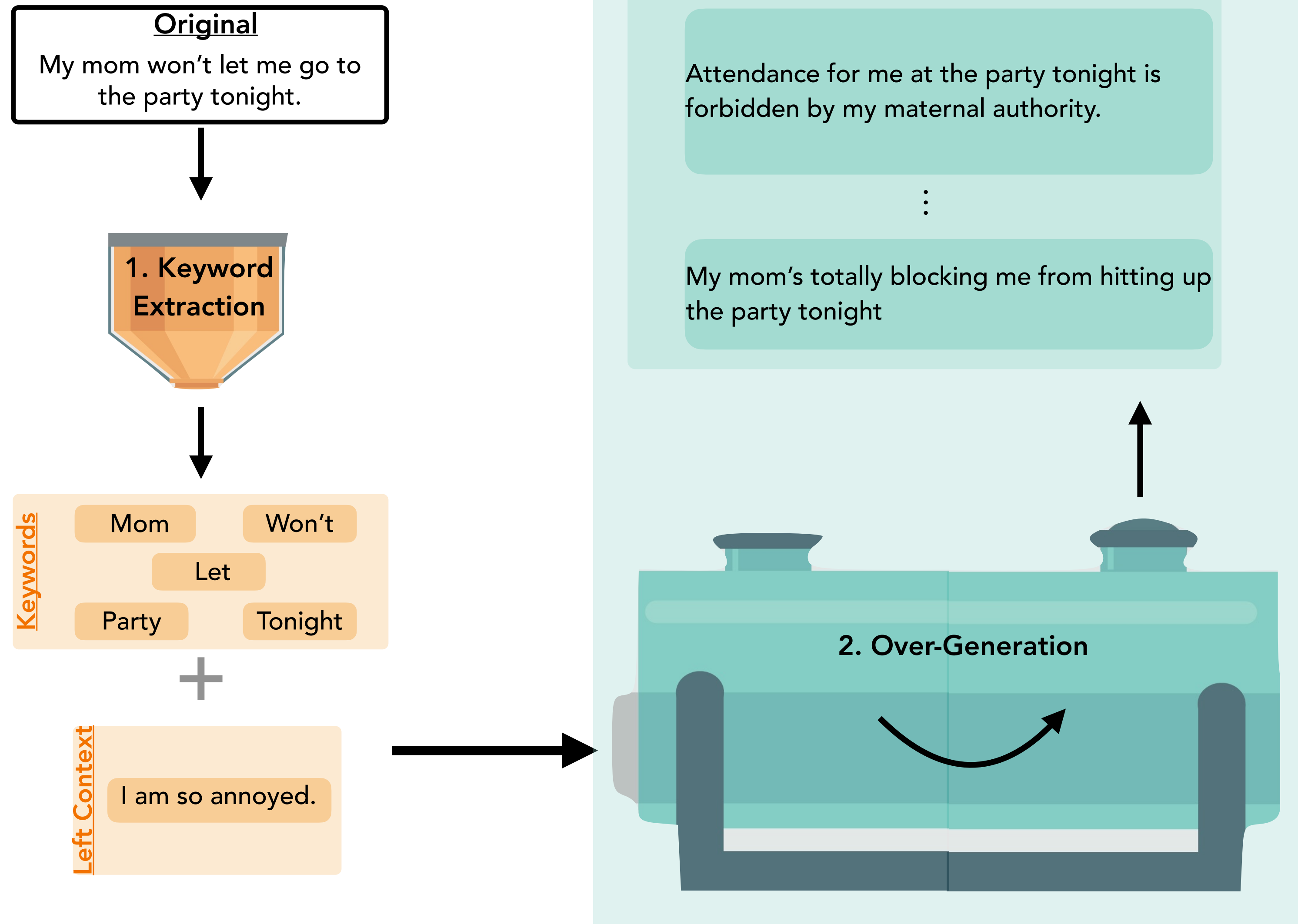
Mom	Won't
Let	
Party	Tonight

+

Left Context

I am so annoyed.

Innovations

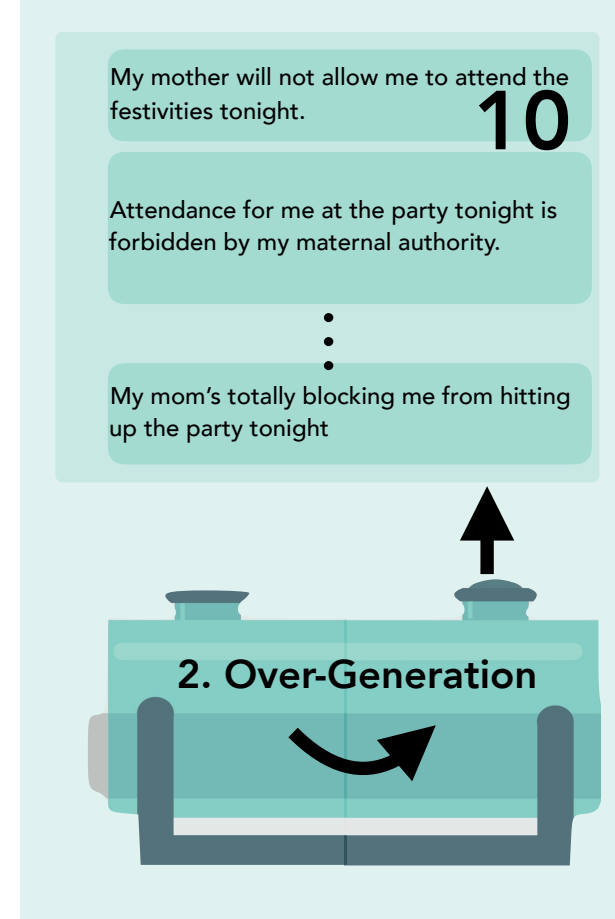


Innovations: Over-Generation

Constrained to
original content

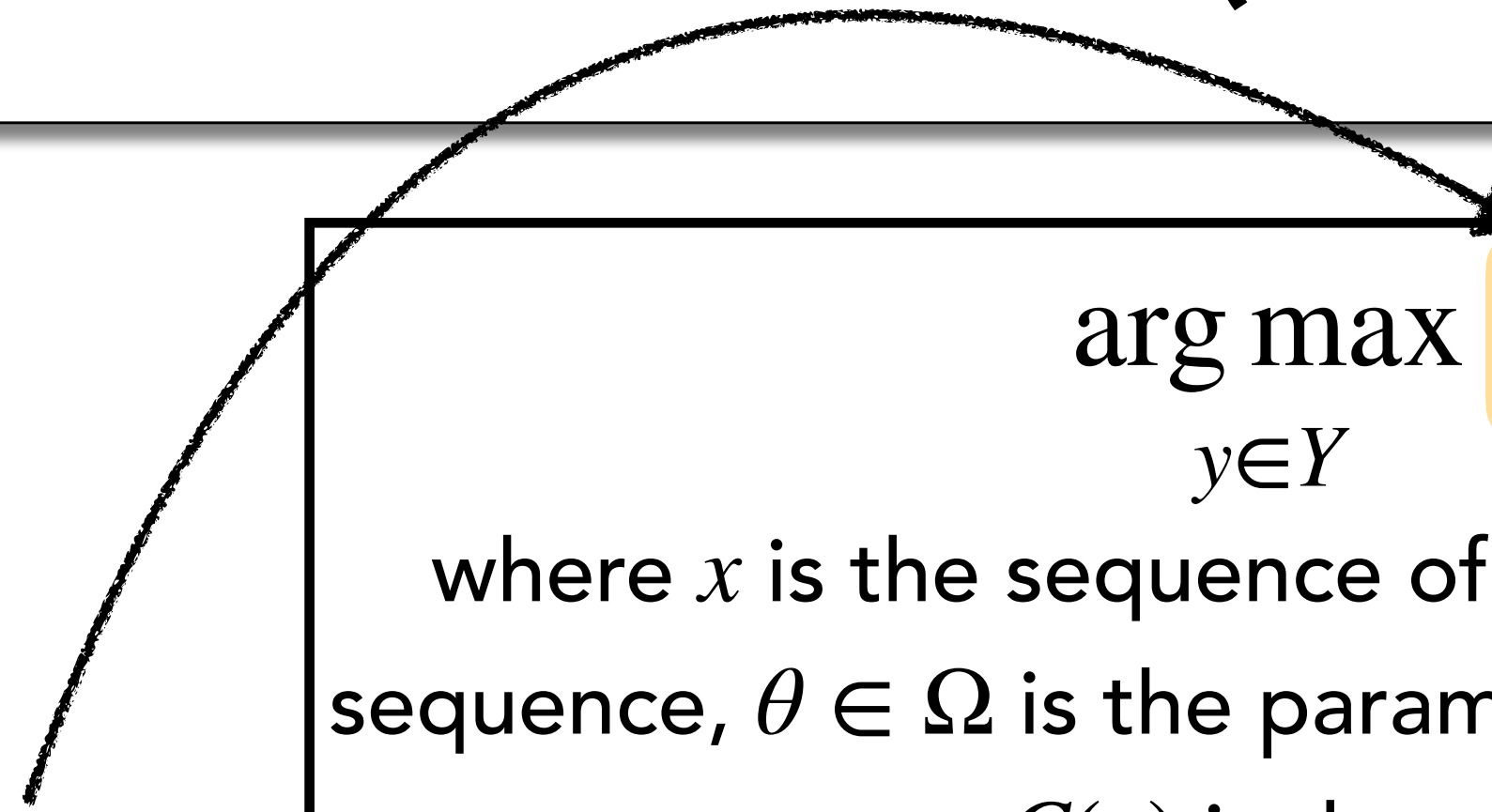


Create diverse
authorship styles



Constrained + Diverse Beam Search
(CoDi-BS)

Constrained + Diverse Beam Search (CoDi-BS)


$$\arg \max_{y \in Y} P_{\theta}(y | x) + \lambda C(y)$$

where x is the sequence of previous tokens, $y \in Y$ is the output sequence, $\theta \in \Omega$ is the parameter vector, λ is a hyperparameter, and $C(y)$ is the constraint penalization.

Add Diversity

$$P^*(y | x) = P_{\theta}(y | x) - \lambda F$$

$F \in \mathbb{R}^v$ is a vector of frequencies of tokens chosen in the previous beams, and λ is a hyperparameter

Innovations: Constrained + Diverse Beam Search¹²

$$P^*(y | x) = P_{\theta}(y | x) - \lambda F$$

$F \in \mathbb{R}^v$ is a vector of frequencies of tokens chosen in the previous beams, and λ is a hyperparameter

Beam	V1	V2	V3	V4	V5
1	0.9	0.8	0.3	0.2	0.7

2	0.8	0.5	0.4	0.8	0.6
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$\lambda \times 0.2 \times$

1	0	0	0	0
---	---	---	---	---

F

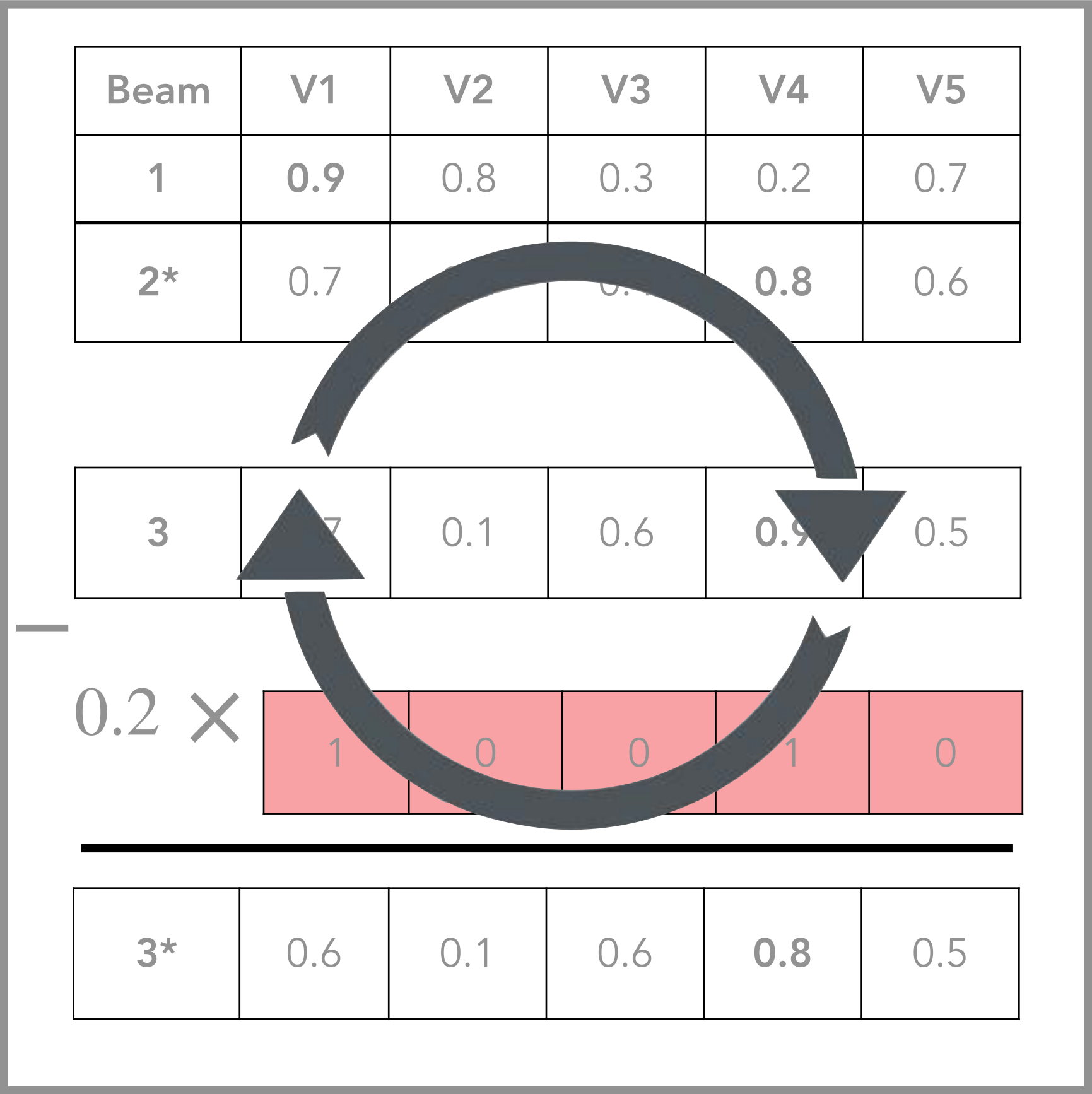
2*	0.7	0.5	0.4	0.8	0.6
----	-----	-----	-----	-----	-----

Replace & Next Beam

Innovations: Constrained + Diverse Beam Search¹³

$$P^*(y|x) = P_{\theta}(y|x) - \lambda F$$

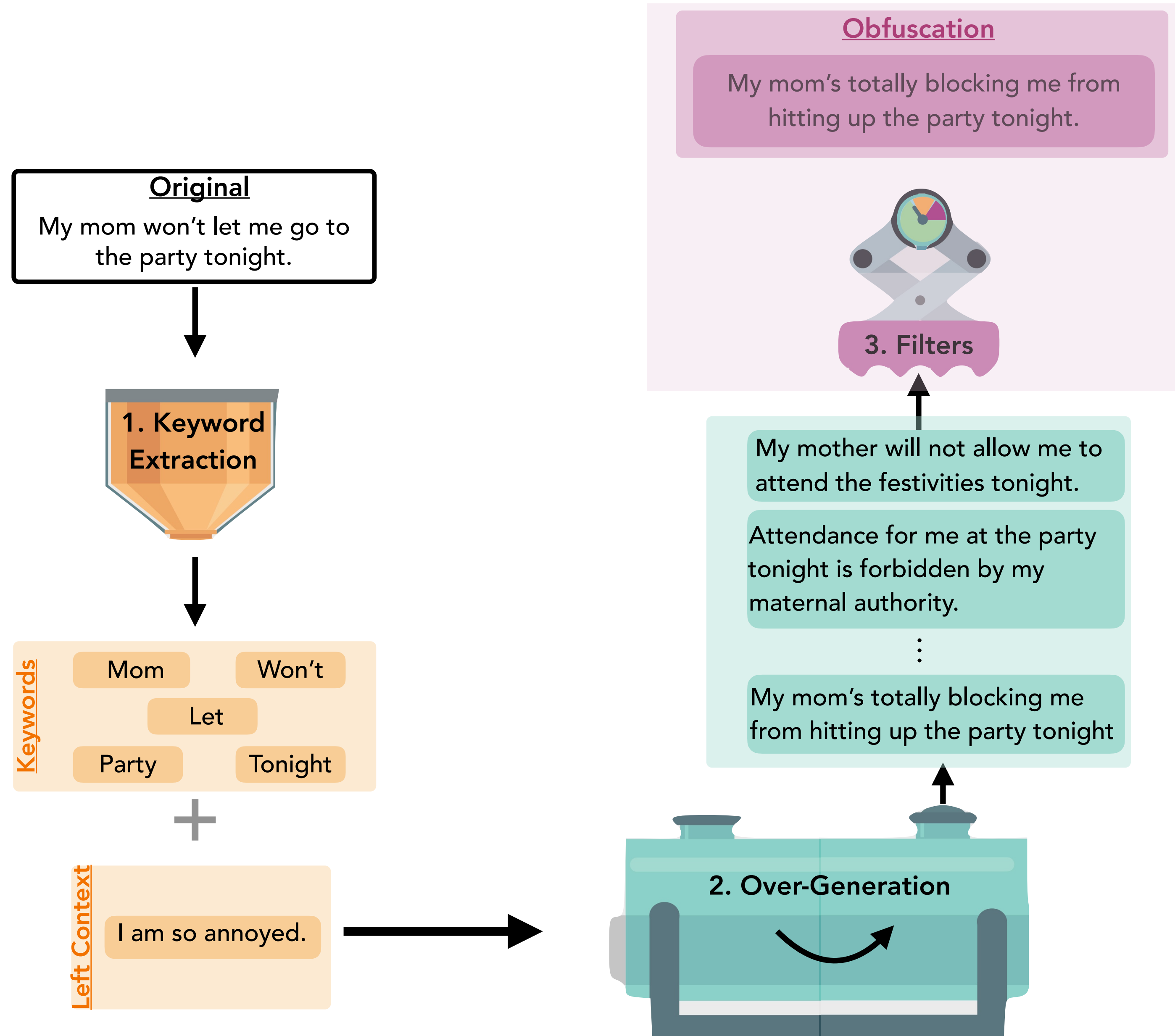
$F \in \mathbb{R}^v$ is a vector of frequencies of tokens chosen in the previous beams, and λ is a hyperparameter



Diversity Processed Logits: $P^*(y x)$					
Beam	V1	V2	V3	V4	V5
1	0.9	0.8	0.3	0.2	0.7
2*	0.7	0.5	0.4	0.8	0.6
3*	0.6	0.1	0.6	0.8	0.5
4*	0.3	0.5	0.7	0.4	0.7

Innovations

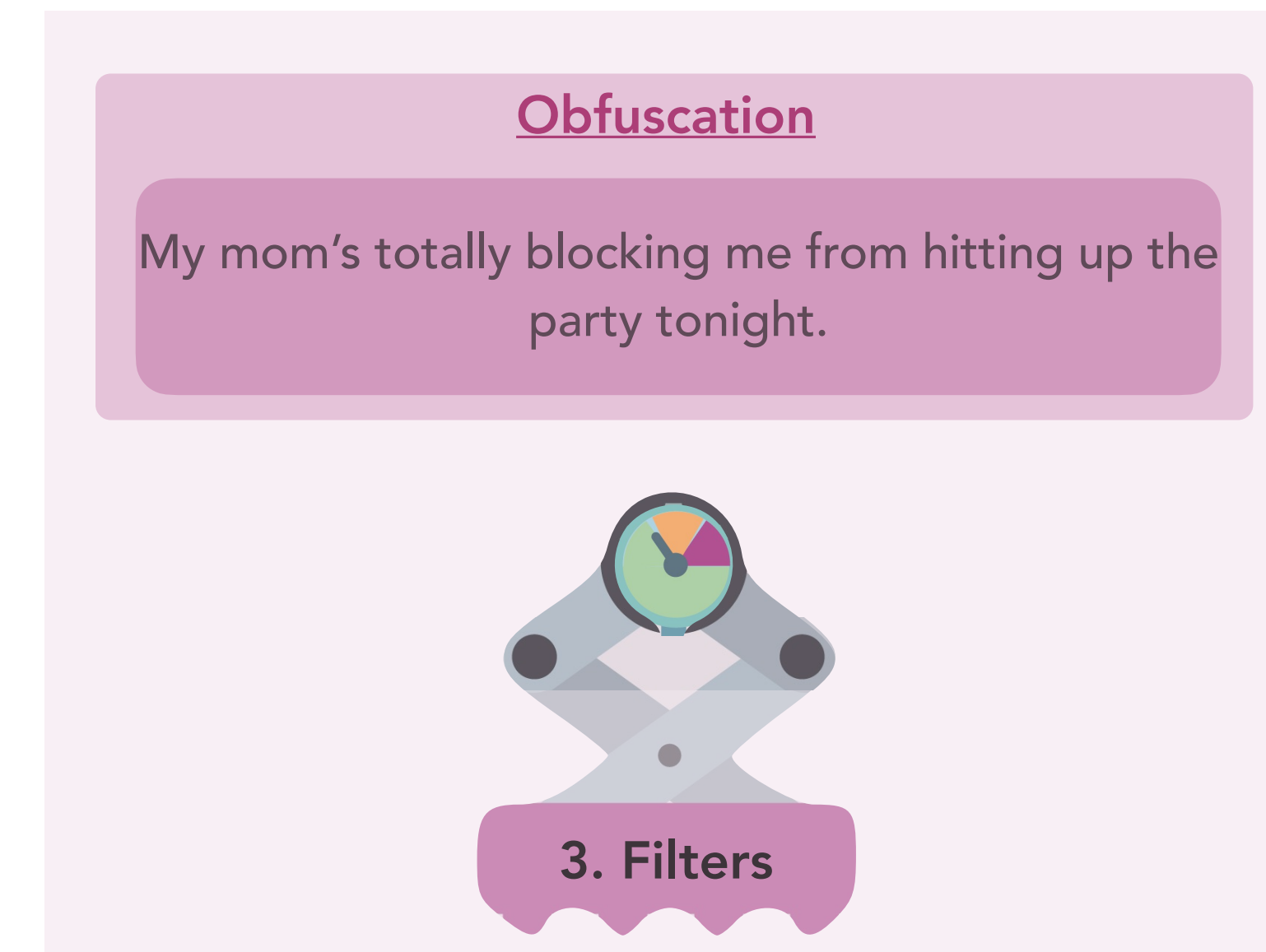
14



Innovations: Filtering

Filtering

- Reduce pool and allow user personalization
- We used the following:
 - Grammar: Corpus of Linguistics Acceptability (CoLA)
 - Content Preservation: Natural Language Inference (NLI)
- Customizable!
 - Length
 - Formality
 - Grade level
 - And more



How does JAMDEC perform compared to other methods?



JAMDEC: Experimental Setup

• Two Datasets

1. Extended-Brennan-Greenstadt: collection of formal scholarly passages
2. Blog Authorship Corpus: diary-style entries from blog.com
 - Number of Authors: 3,5, or 10



• Baselines

- *Stylometric*: rule-based changes such as synonyms, number of words, punctuation, etc.
 - *Round Trip Machine Translation*: English —> German —> French —> English
 - Paraphrase
 - *Mutant-X*: Iteratively re-writes and combines randomly, uses internal classifier
-
- Base Model: GPT2-XL (1.5B)

JAMDEC: Evaluation Metrics



- Authorship obfuscation is traditionally evaluated (automatically) on:

1. Obfuscation

How well does the rewritten text obfuscate the author style?

Metric: *Drop-Rate* using automatic authorship classifier (ENS and BertAA)

2. Fluency

How understandable is the text?

Metric: *Probability of acceptable grammar* using CoLA model

3. Content Preservation

How similar in meaning is the generation to the original text?

Metric: *Probability of two-way entailment* using NLI model

- Overall Task Score: **average** of the three metrics

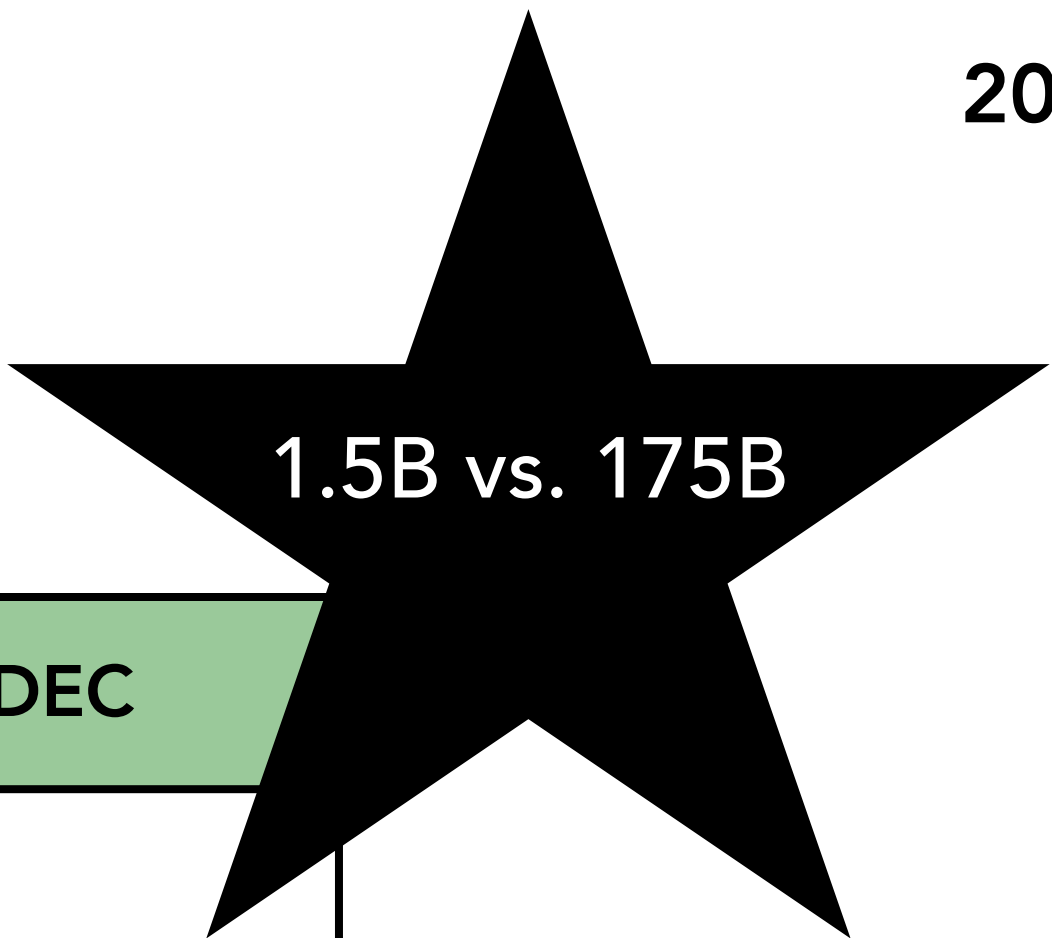
$$\text{Task Score} = \frac{\text{Drop Rate} + \text{NLI} + \text{CoLA}}{3}$$

JAMDEC: Automatic Evaluation

Dataset	Metric	Mutant-X	Paraphrase	Machine	Stylometric	JAMDEC
Scholar - 3	Drop Rate (ENS) ↑	-0.04	0.04	0.04	-0.03	0.11
	Drop Rate (BertAA) ↑	0.04	0.04	0.08	0.12	0.04
	NLI ↑	0.61	0.62	0.75	0.50	0.81
	CoLA ↑	0.51	0.78	0.69	0.46	0.79
	Task Score (ENS) ↑	0.36	0.48	0.49	0.31	0.57
	Task Score (BertAA) ↑	0.39	0.48	0.51	0.36	0.55
Scholar - 5	Drop Rate (ENS)	0.08	0.2	0.2	0.23	0.13
	Drop Rate (BertAA)	0	-0.06	0.07	0.04	0.14
	NLI	0.57	0.62	0.74	0.48	0.82
	CoLA	0.55	0.77	0.69	0.46	0.79
	Task Score (ENS)	0.4	0.53	0.54	0.39	0.58
	Task Score (BertAA)	0.37	0.44	0.50	0.33	0.58
Blog - 10	Drop Rate (ENS)	0.13	0.35	0.3	0.21	0.32
	Drop Rate (BertAA)	0.06	0.4	0.11	0.08	0.32
	NLI	0.61	0.46	0.62	0.75	0.67
	CoLA	0.45	0.62	0.54	0.41	0.74
	Task Score (ENS)	0.4	0.48	0.49	0.46	0.58
	Task Score (BertAA)	0.37	0.49	0.42	0.41	0.58

JAMDEC
had the
highest
overall Task
Score on
every
dataset!

JAMDEC: Automatic Results



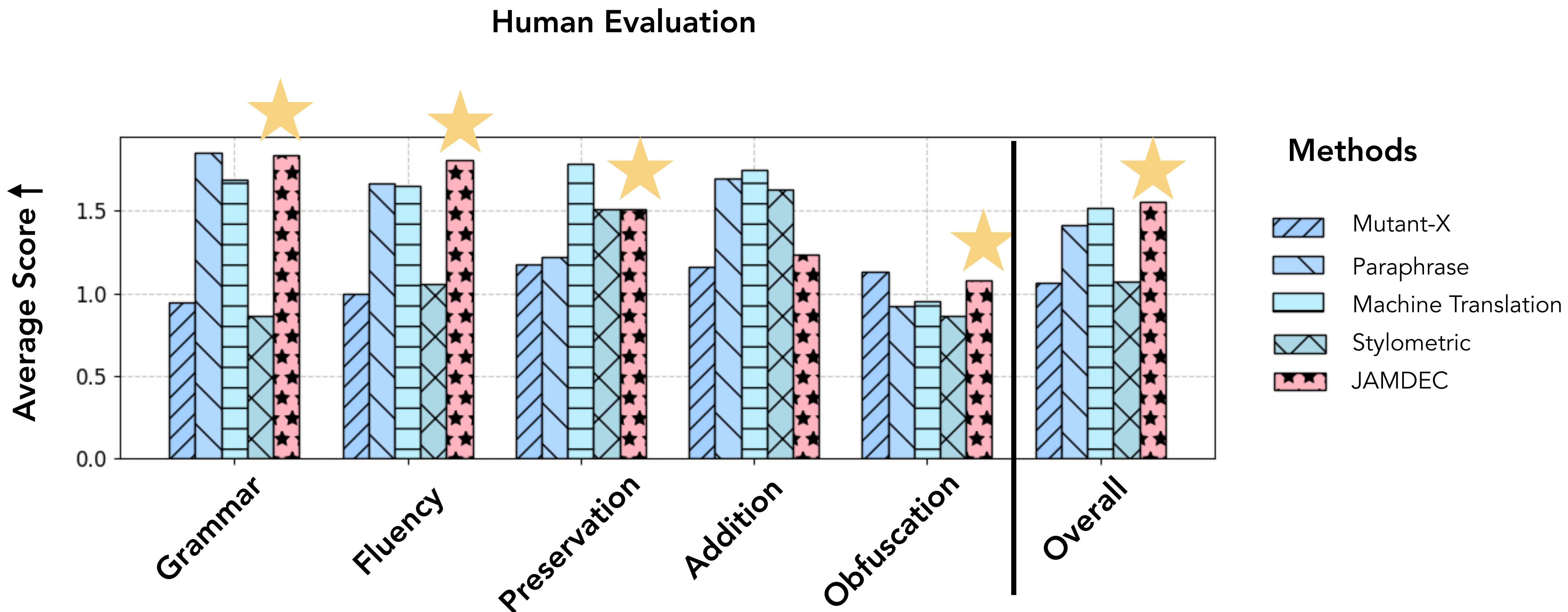
		GPT3-Turbo		JAMDEC
Dataset	Metric	Sentence	Paragraph	
Scholar - 3	Drop Rate (ENS) ↑	0.23	0.23	0.11
	Drop Rate (BertAA) ↑	0.13	0.09	0.04
	NLI ↑	0.77	0.73	0.81
	CoLA ↑	0.76	0.8	0.79
	Task Score (ENS) ↑	0.59	0.59	0.57
	Task Score (BertAA) ↑	0.55	0.54	0.55

Performs similarly to much larger models!

Do humans agree that JAMDEC outperforms other methods?



JAMDEC: Human Evaluation



*3-point Likert Scale by 3 raters each

JAMDEC: Qualitative Results

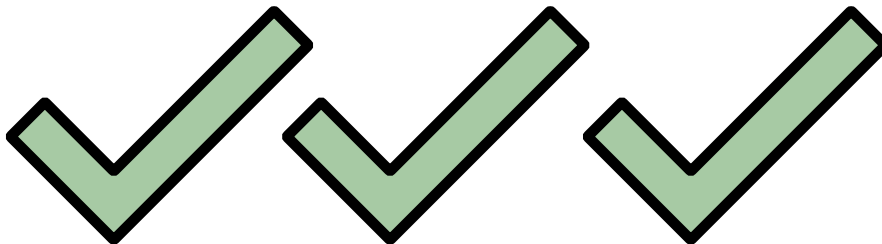
Method	Generation
Original	The Ex. An ex holding a grudge can do a lot of damage in a short amount of time. He knows enough to open accounts in your name, and he has the motive to hurt you.
Mutant-X	The Ex. An ex holding a bitterness able ought a lot of damage in a length quantity of time. He knows enough to ascend accounts in Your prefix , and he has the justifiable to impair You .
Paraphrase	A lot of damage can be done In a short period of time. He knows how to open accounts In your name and he wants to hurt you.
Machine Translation	The former. An old man who holds a knife can make a lot of damage in a short time. He knows enough to open accounts in your name, and he has the reason to hurt you.
Stylometric	An ex holding, a grudge can do a lot inside damage in a brief amount in time, yet he knows enough to open accounts in your name, and he has the motive to hurt you.
JAMDEC	The Ex. When the ex is holding his grudge against the person who caused him lot of damage to his life, he is short sighted and will do anything in his power to get back at that person, no matter how much it will hurt the person he is trying to get revenge against. He knows enough to open accounts in your name, and he has the motive to hurt you.

Not Grammatical

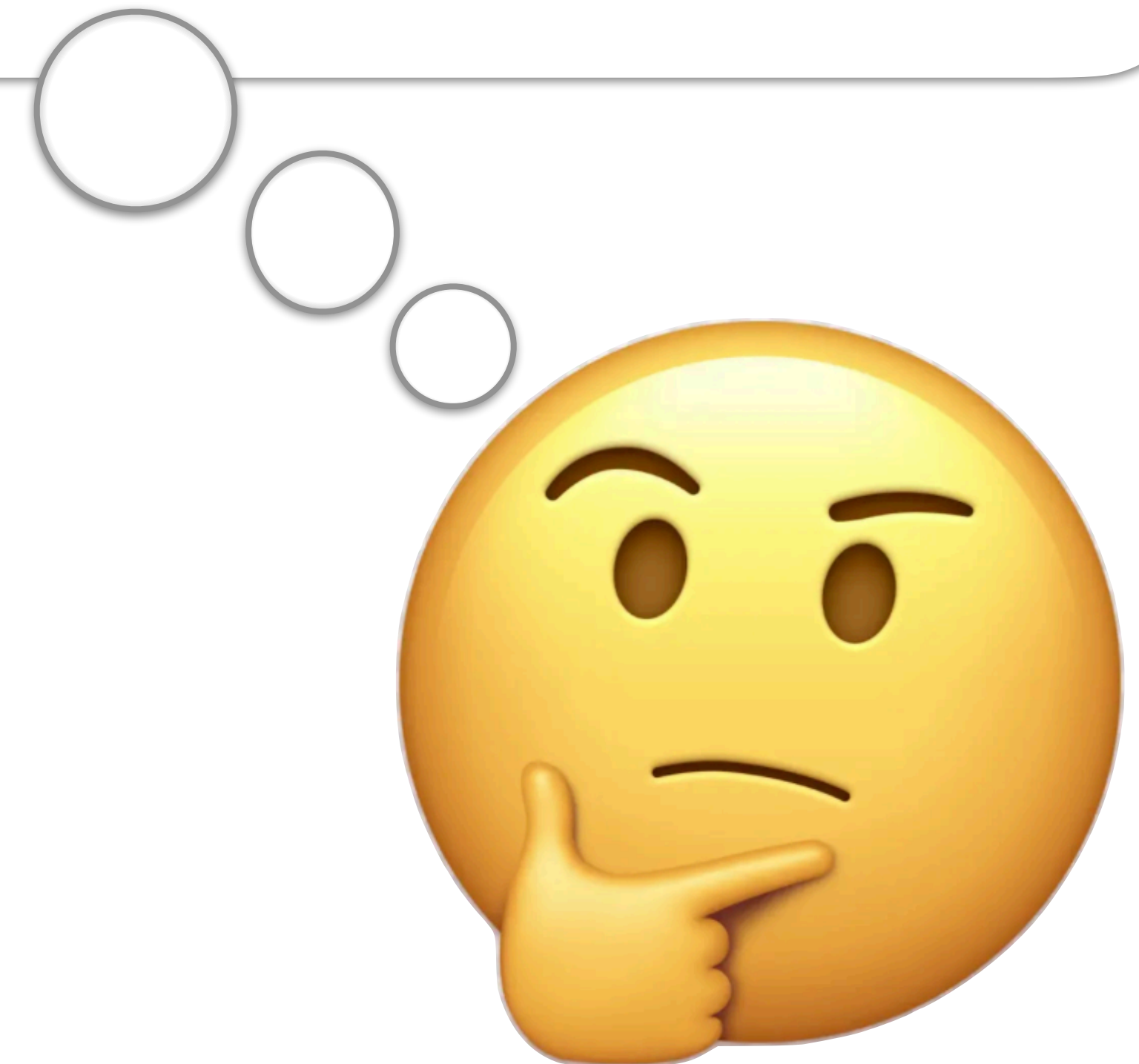
Incorrect Content

Incorrect Content

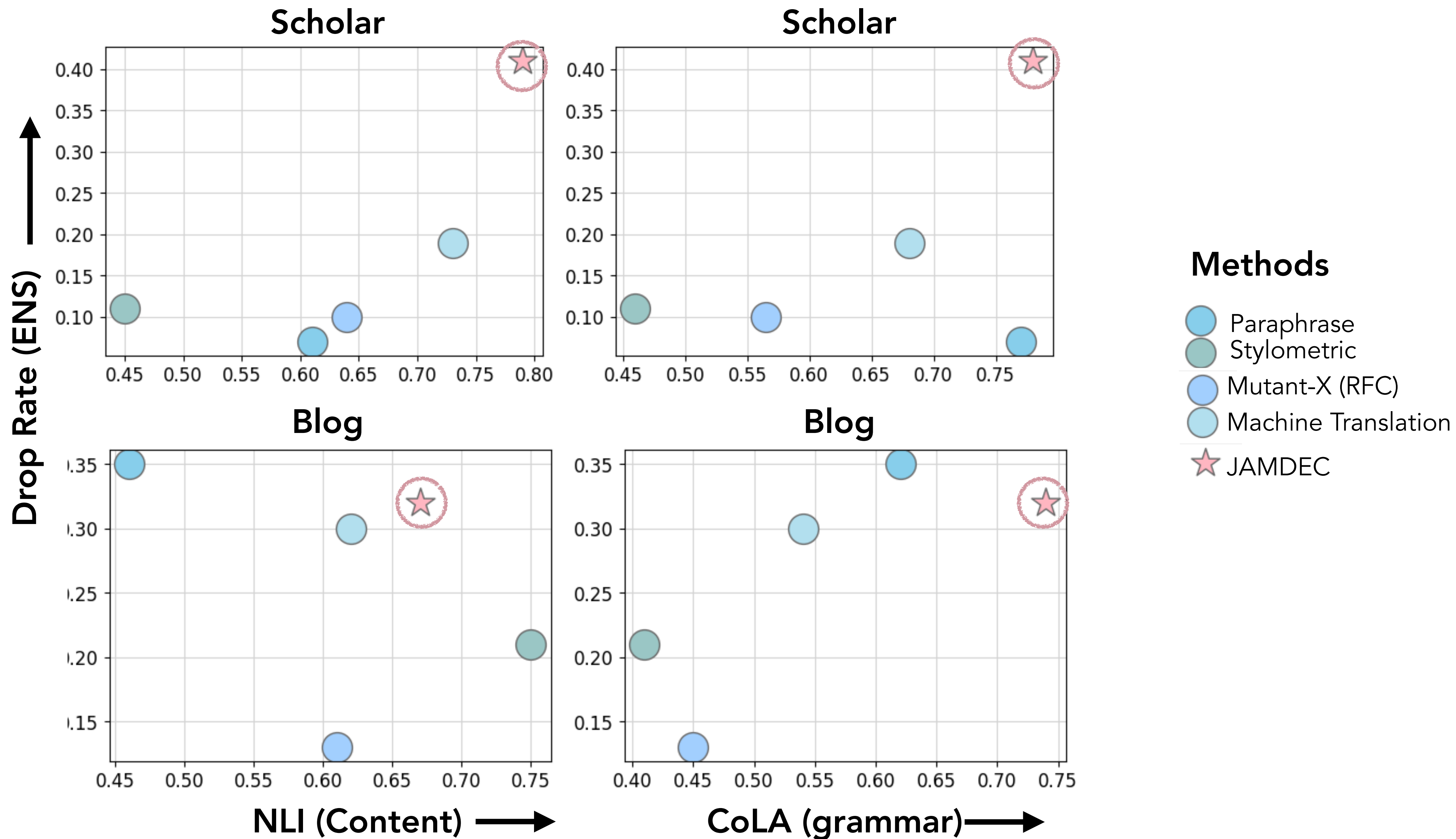
Missing Meaning



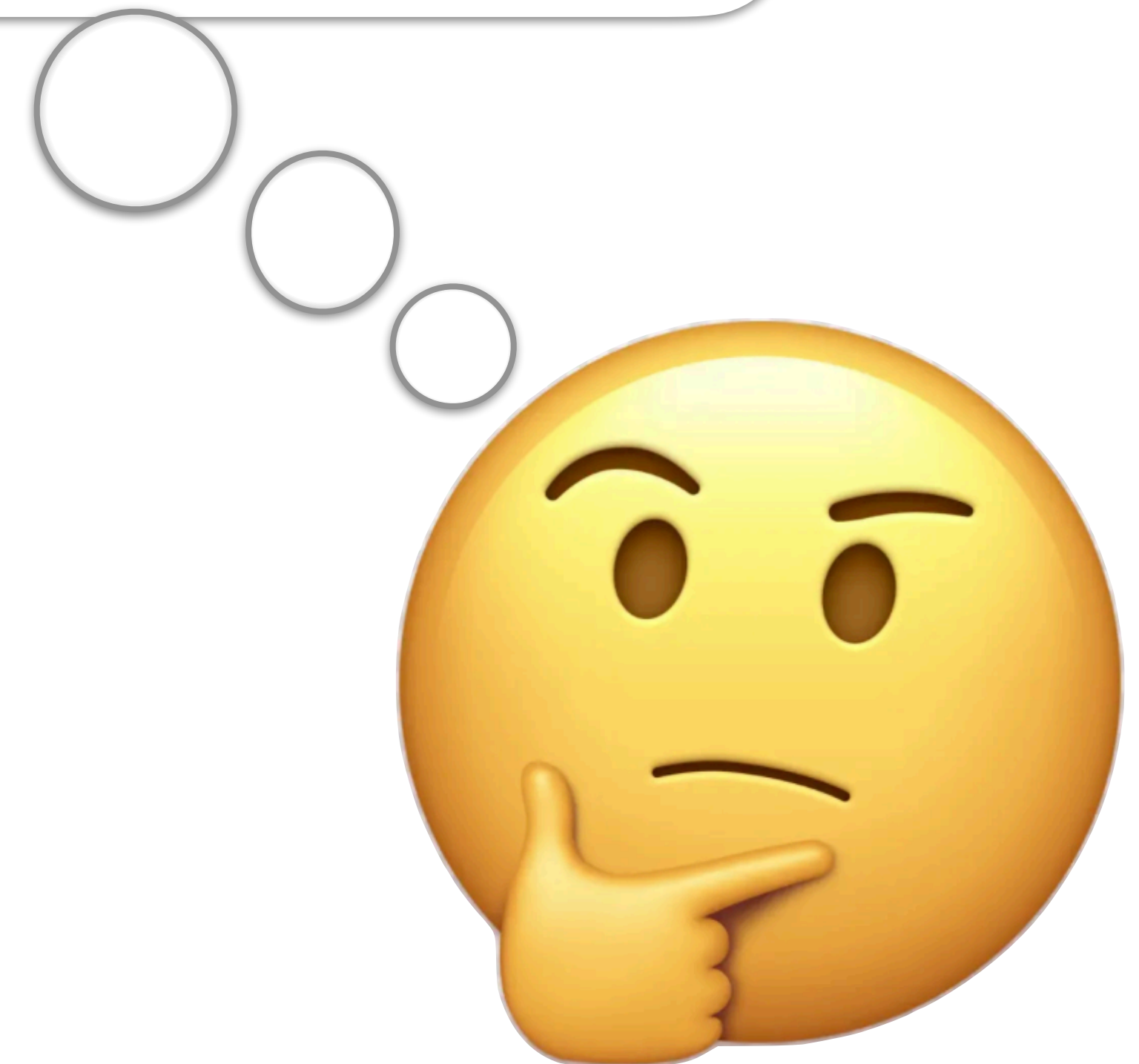
It seems like there might be a tradeoff between obfuscation, content preservation, and fluency...



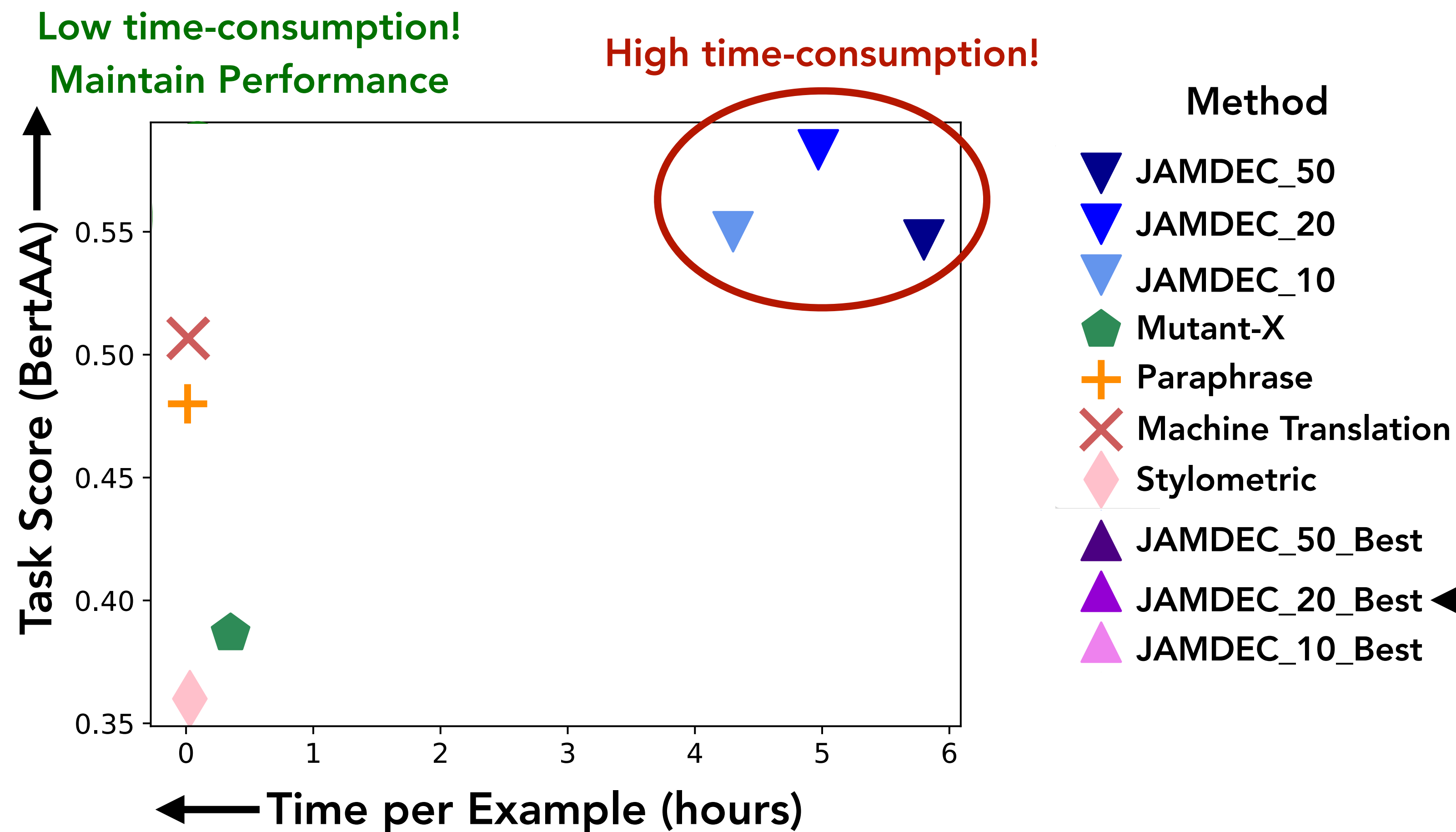
JAMDEC: Inherent Tradeoff



Over-generation seems like it would take more time than other methods...



JAMDEC: Computational Time



Best keyword extraction,
type of constraints, type
of algorithm (sampling?
Diversity?)

JAMDEC

Small Models
($<2B$)

Beats Models 100X
Larger

Customizable

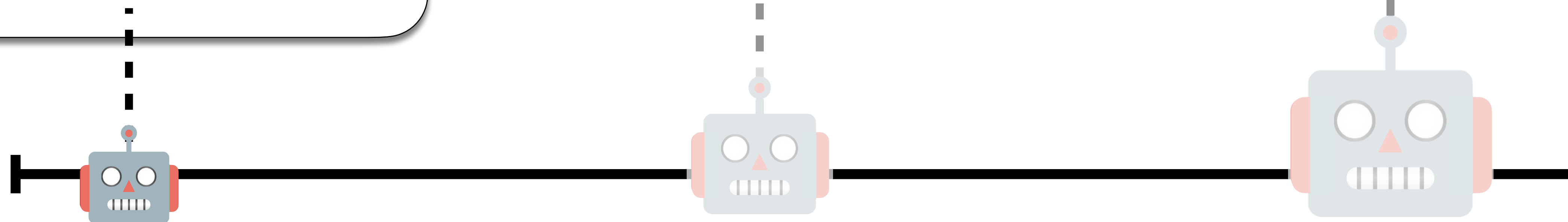
No Training (or Extra
Corpus)

Controllable Generation

Small Models
($<2B$)

Medium Models
($2B - 10B$)

Large Models
($>10B$)



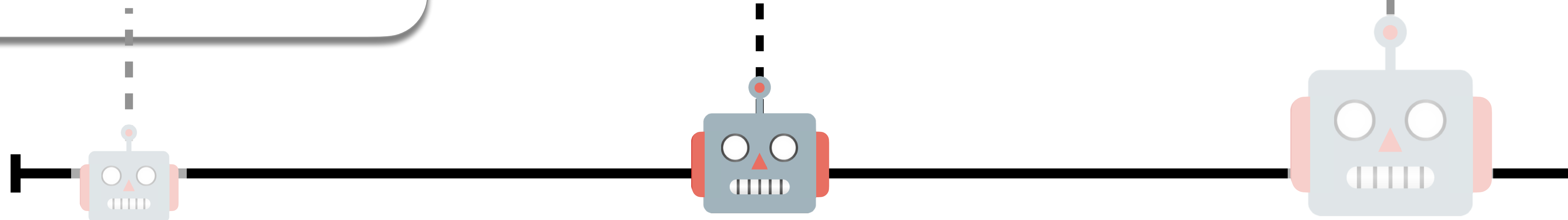
Model Size

Controllable Generation

Open Small Models
($<2B$)

Medium Models
($2B - 10B$)

Closed Large Models
($>10B$)



Model Size

Medium Models (2B - 10B)

Can we add more
control and
interpretability?

Pros

Access to internal
data (e.g. logits)

Higher fluency
capabilities

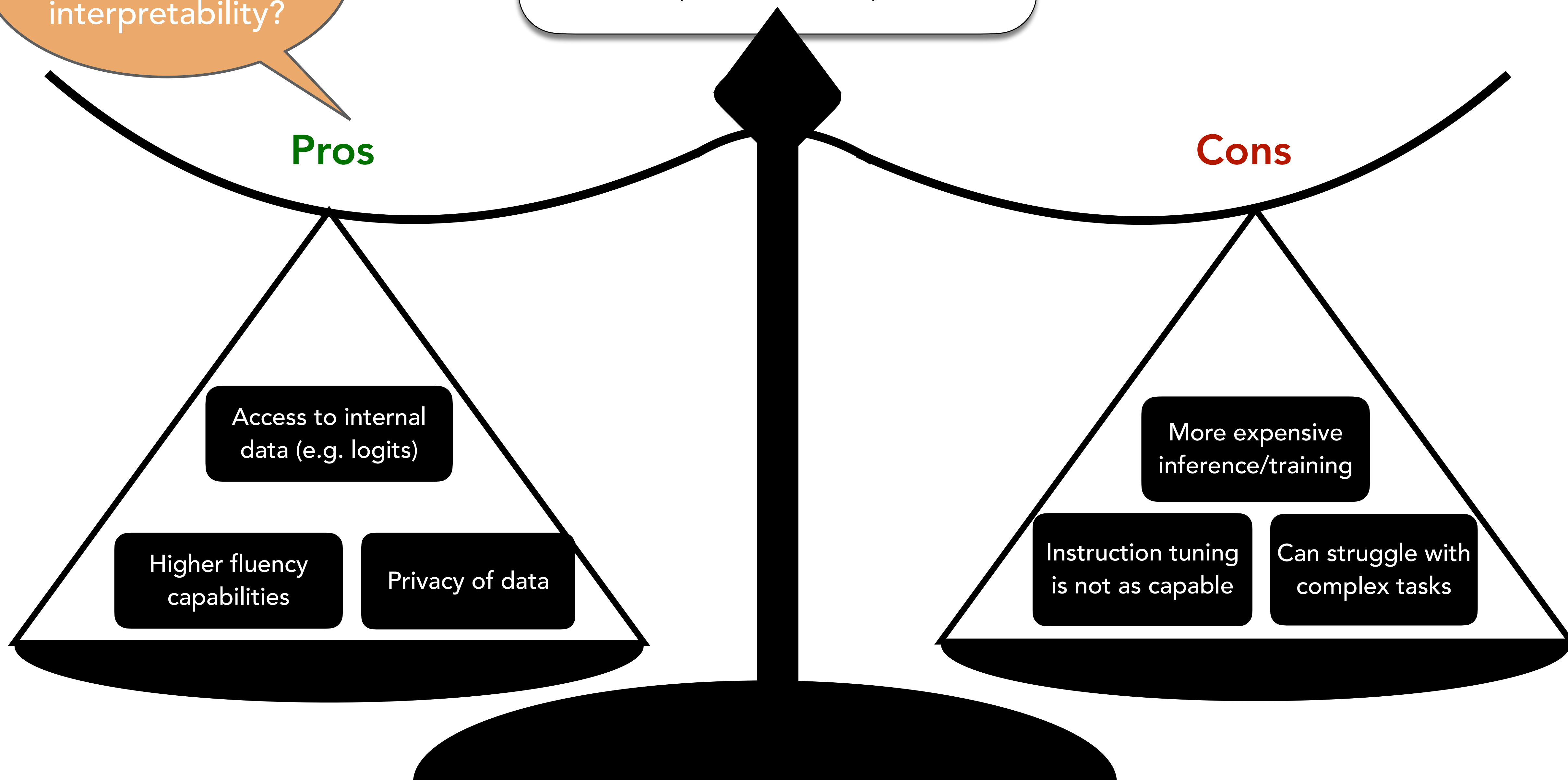
Privacy of data

Cons

More expensive
inference/training

Instruction tuning
is not as capable

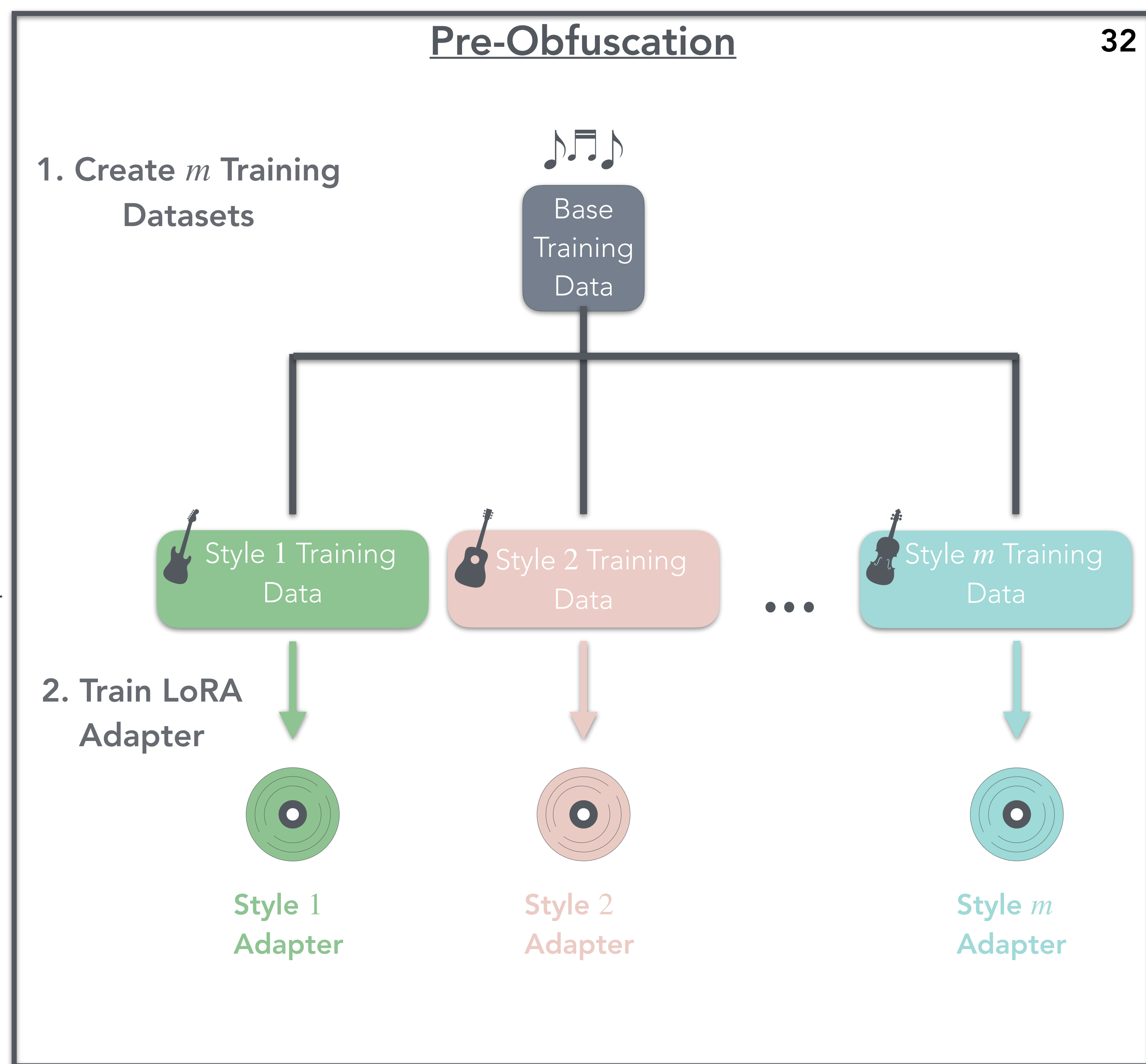
Can struggle with
complex tasks



StyleRemix

Contributions:

- New interpretable obfuscation method
 - 2 datasets: AUTHORMIX + DiSC
 - 16 new style LoRA adapters
 - 3 new style classifiers
-
- An adaptive and interpretable obfuscation method that perturbs specific, fine-grained style elements of the original input text
-
- **Pre-Obfuscation:**
 1. *Generate Training Data for each m style*
 2. *Train Low-Rank Adapters (LoRA Adapter)*



StyleRemix

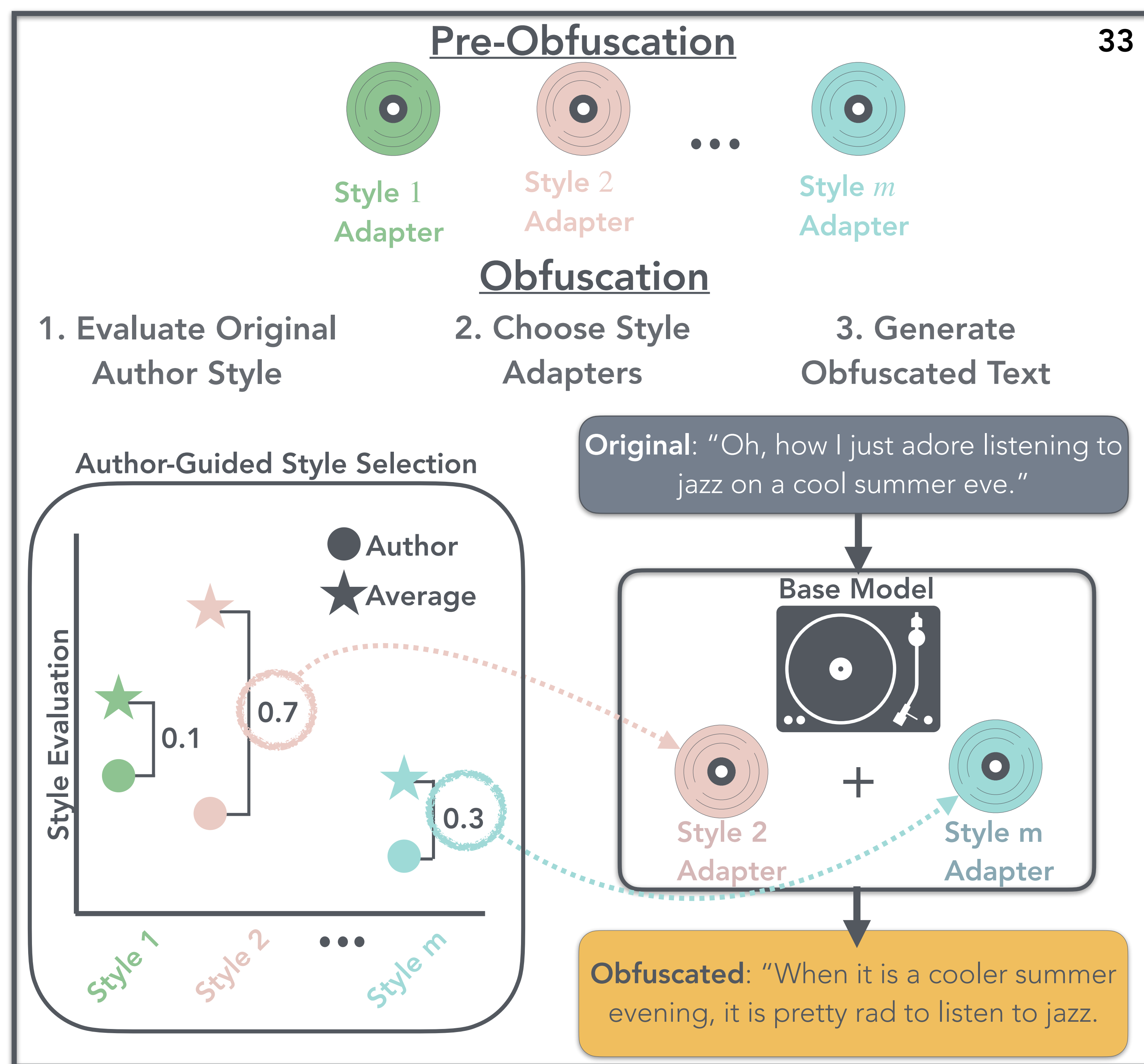
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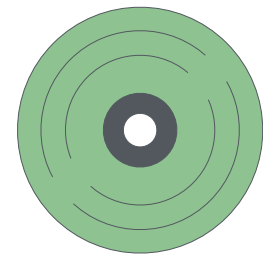
- An adaptive and interpretable obfuscation method that perturbs specific, fine-grained style elements of the original input text

• Obfuscation

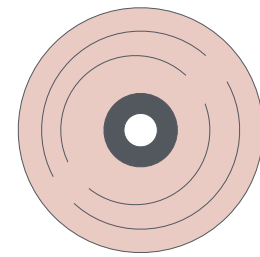
1. Evaluate Original Author Style
2. Choose Style Adapters
3. Generate Obfuscated Text



Pre-Obfuscation

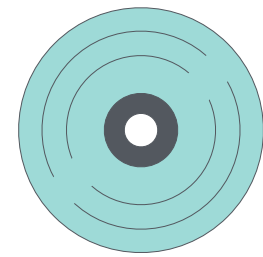


Style 1
Adapter



Style 2
Adapter

...



Style m
Adapter

Which style axes should we use?

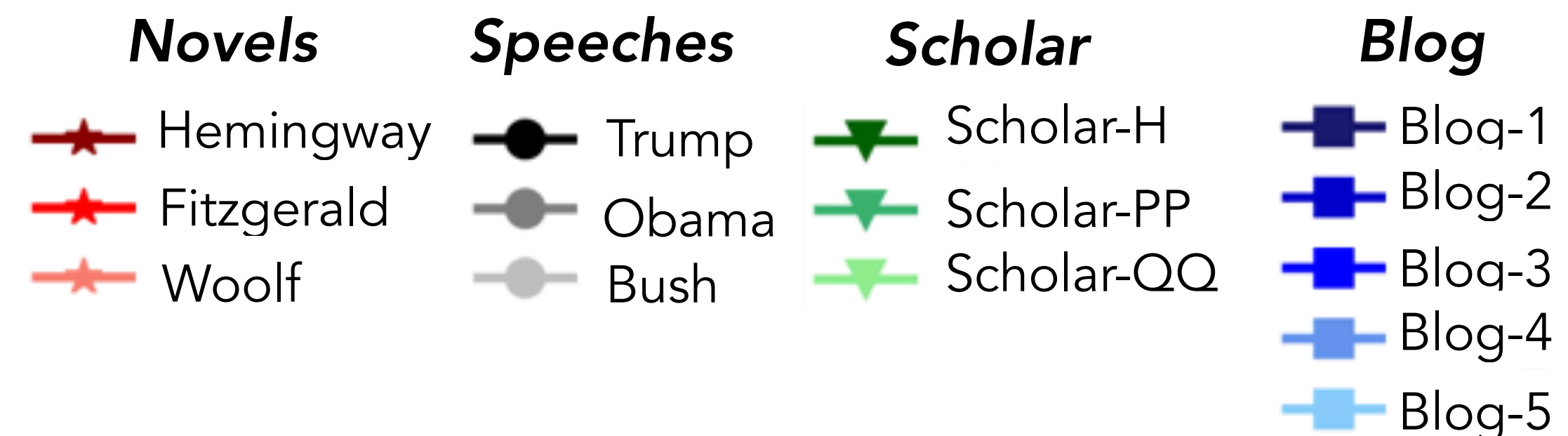
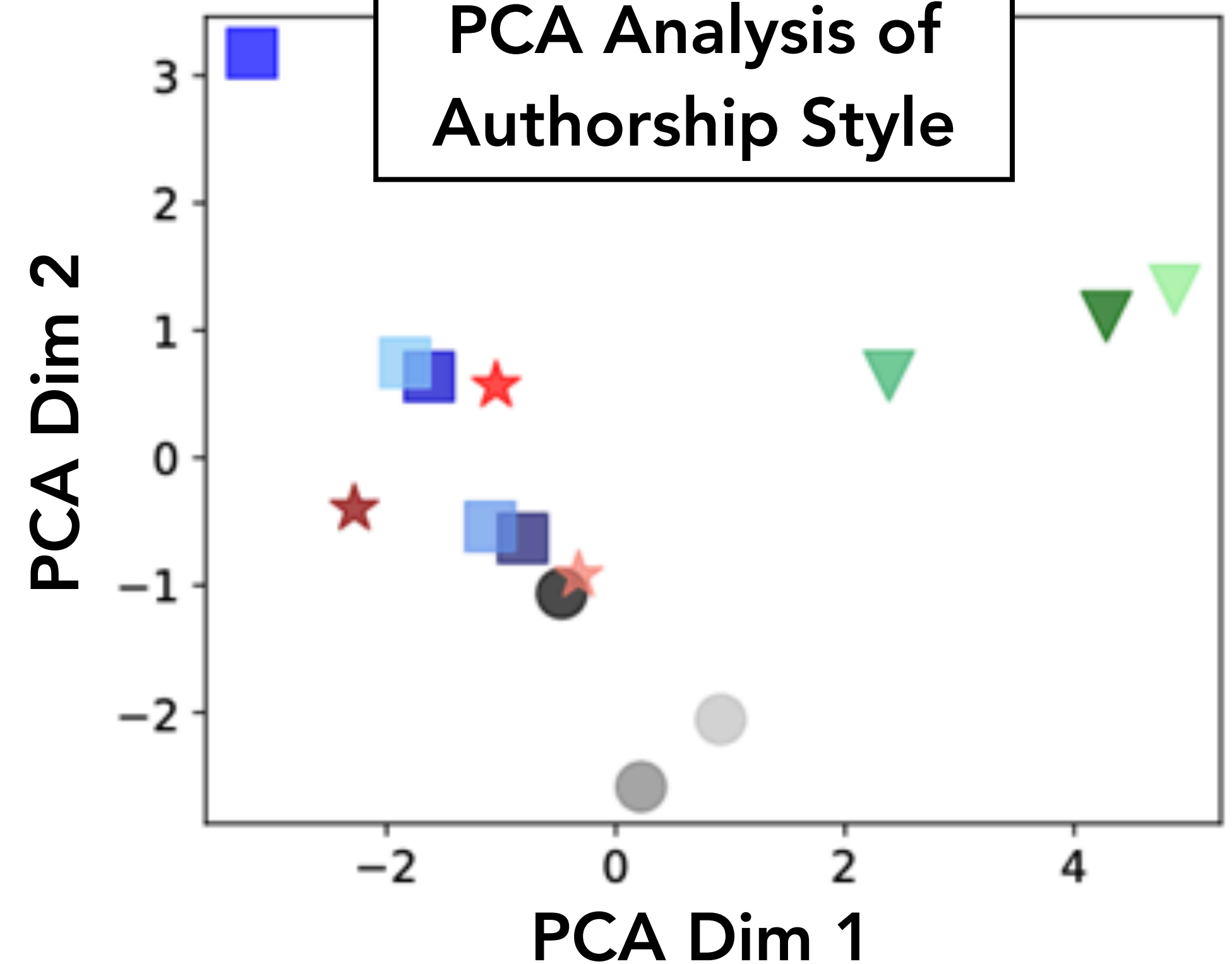
Stylometry

NLP



Do these styles differentiate³⁴ authors?

PCA Analysis of Authorship Style



Pre-Obfuscation: Adapter Training Set

Style Axes

Length	Sarcasm
Function Words	Voice
Grade Level	Writing Intent
Formality	

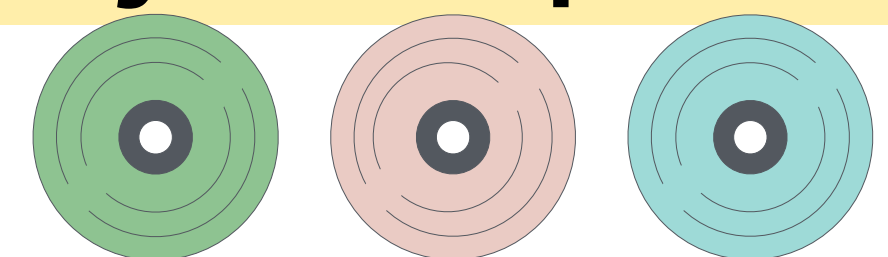
Base Training Dataset

Wikipedia	Books + Plays	Blog
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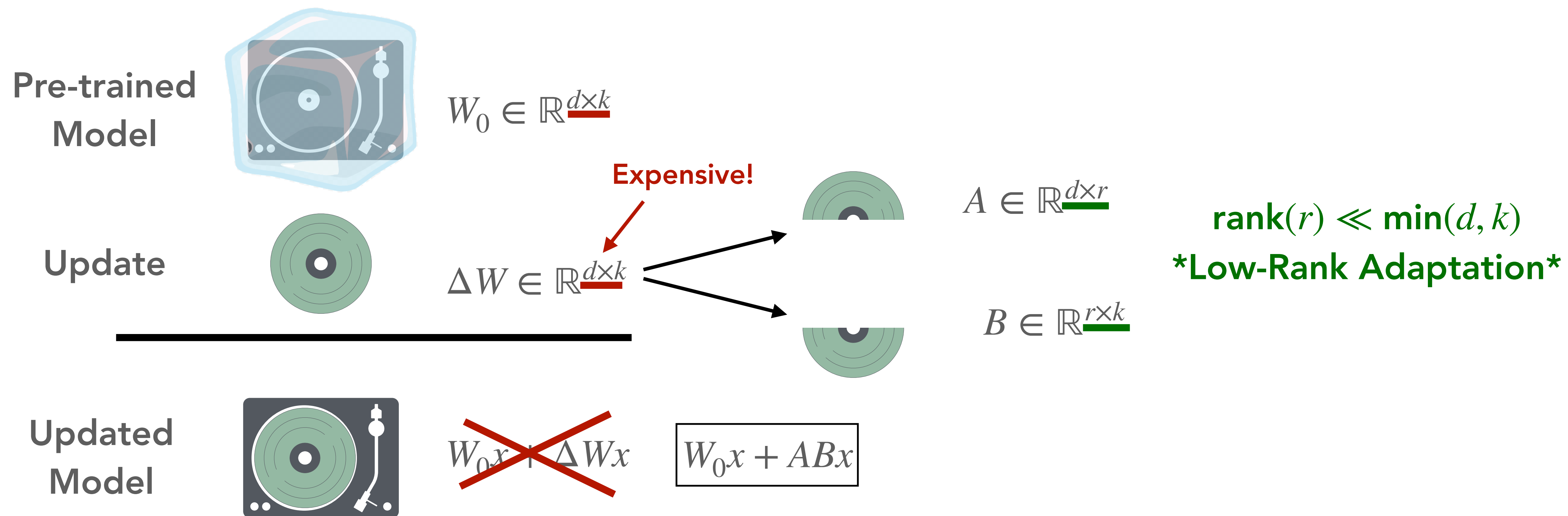
Distilled Style Components Dataset (DiSC)

- A set of web, book, and blog texts rewritten towards **16 distinct style** directions across seven style axes

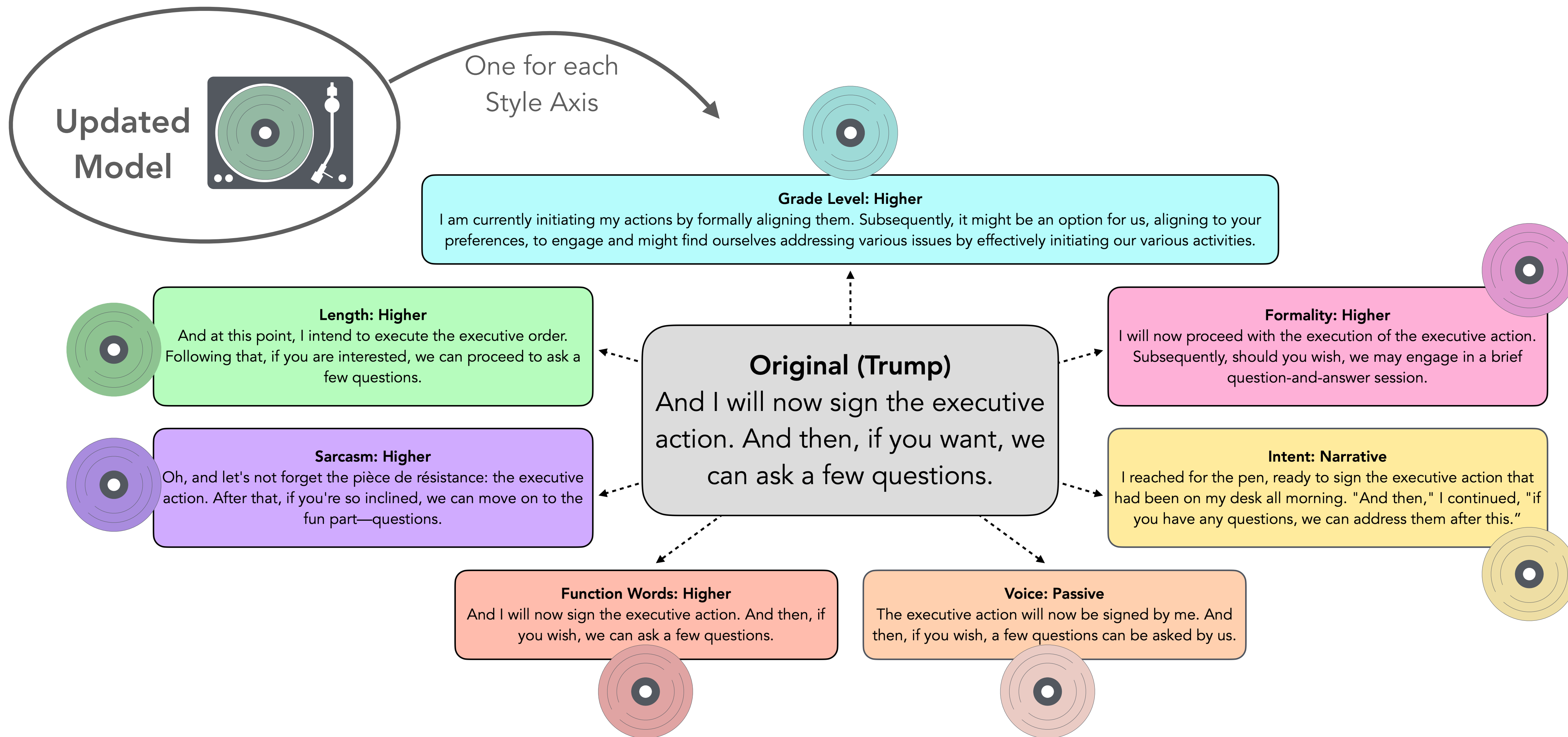
Used to train style adapters!



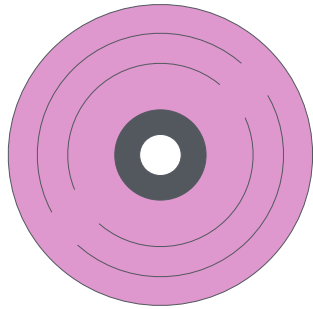
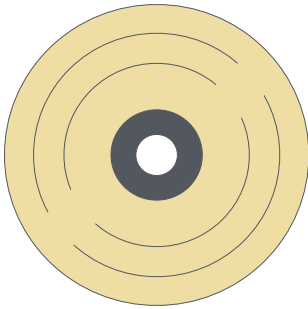
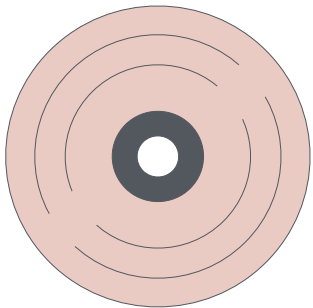
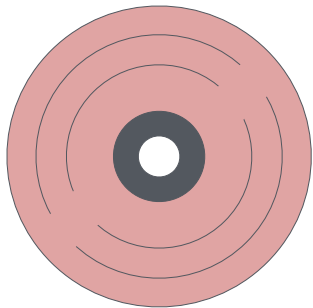
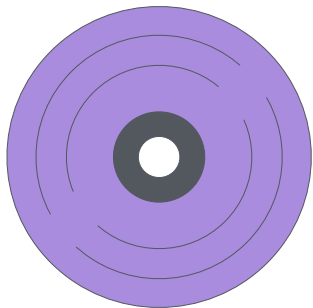
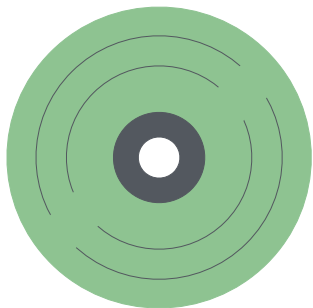
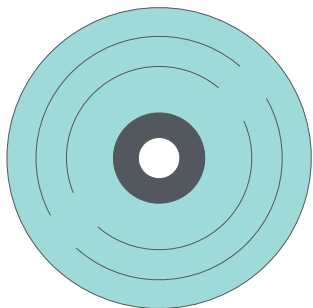
Pre-Obfuscation: Train LoRA Adapter



Pre-Obfuscation: Train LoRA Adapter



Pre-Obfuscation: Train LoRA Adapter



Style Axis (<i>metric</i>)	Original	More	Less
Length (<i>words/sent</i>)	18.87	23.04	<u>18.24</u>
Function Words (<i># func. words</i>)	40.08	55.19	<u>21.47</u>
Grade Level (<i>avg. of 3</i>)	9.45	11.08	<u>6.72</u>
Formality (<i>model score</i>)	0.68	0.97	<u>0.43</u>
	Accuracy (<i>human evaluation</i>)		
Sarcasm	97.7		
Voice	93.7		
Writing Intent (<i>4 classes</i>)	77.7		

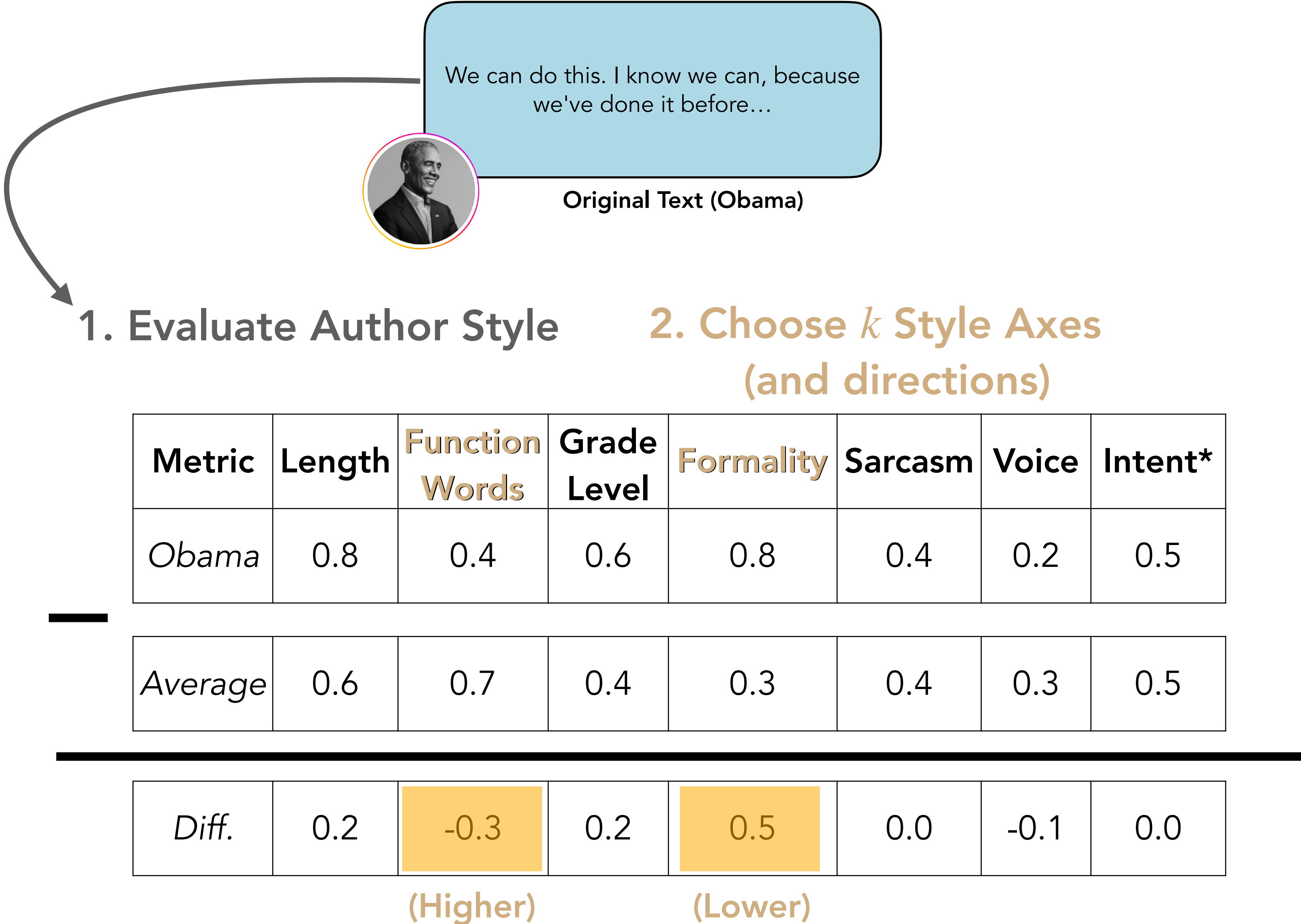
How do we select the LoRA adapters?

We can do this. I know we can, because
we've done it before...

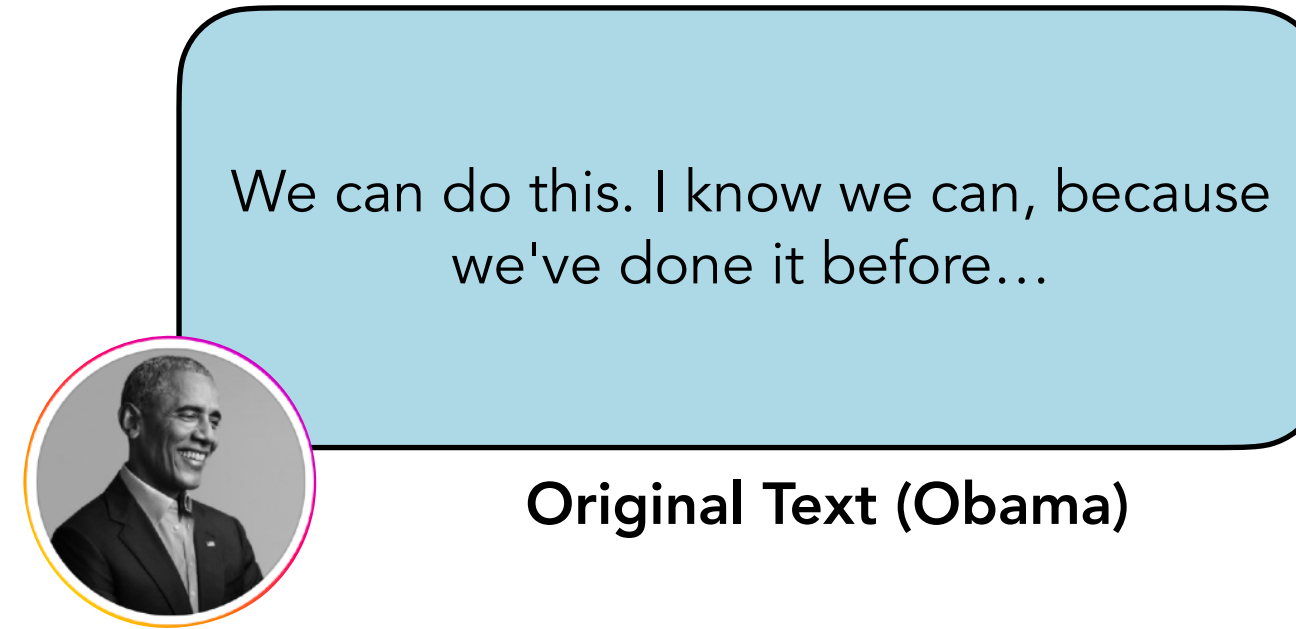


Original Text (Obama)

Obfuscation: Select Style Axes



Obfuscation: Select Style Axes Weights



1. Evaluate Author Style

2. Choose k Style Axis
(and directions)

Function Words (Higher)	Formality (Lower)
----------------------------	----------------------



3. Choose Weights of
Style Axes

3.a) Static Weight Selection

of std. from the average: $\text{std}(\bar{x}_i)$

$$w_i = \begin{cases} 0.7, & \text{if } \text{std}(\bar{x}_i) \leq 1 \\ 0.9, & \text{if } 1 < \text{std}(\bar{x}_i) \leq 2 \\ 1.2, & \text{if } 2 < \text{std}(\bar{x}_i) \leq 3 \\ 1.5, & \text{if } \text{std}(\bar{x}_i) > 3 \end{cases}$$

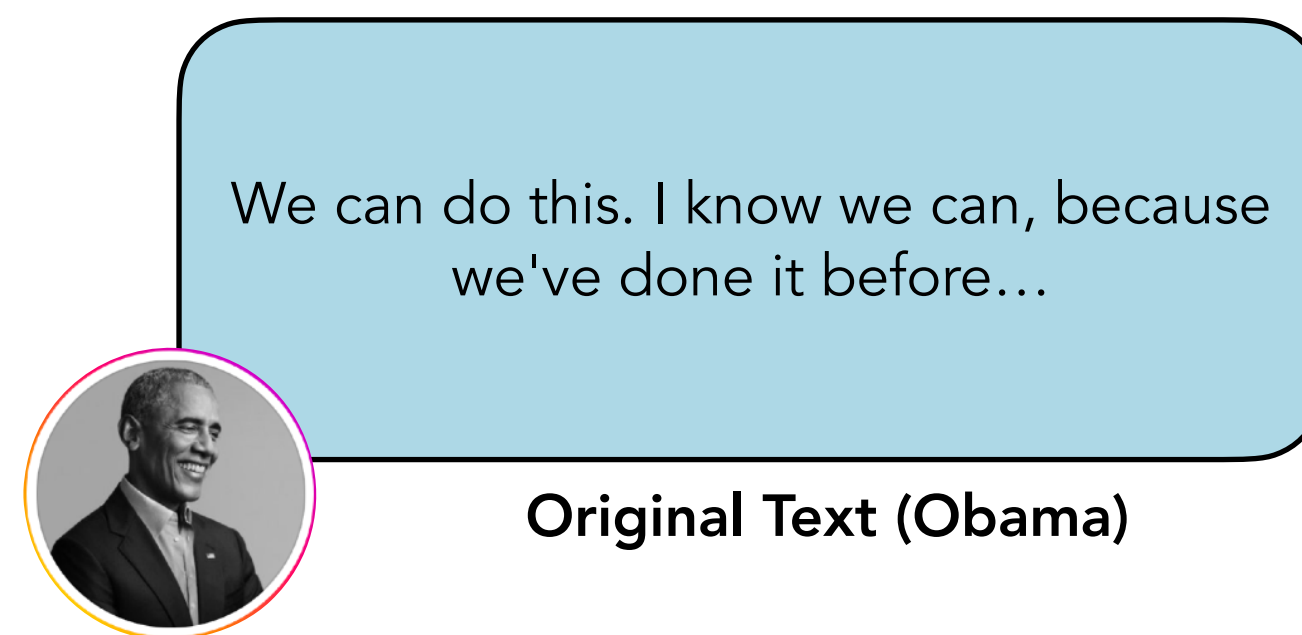
3.b) Dynamic Weight Selection

Optimization of loss based on style axes evaluations

$$L = \sum_{v_i \in \{v_1, v_2\}} \begin{cases} v_i, & \text{if higher} \\ 1 - v_i, & \text{if lower} \end{cases} + \alpha \cdot f$$

v_i = Average style score on test set f = Fluency score

Obfuscation: Select Style Axes Merging



1. Evaluate Author Style

2. Choose k Style Axis
(and directions)

Function Words
(Higher)

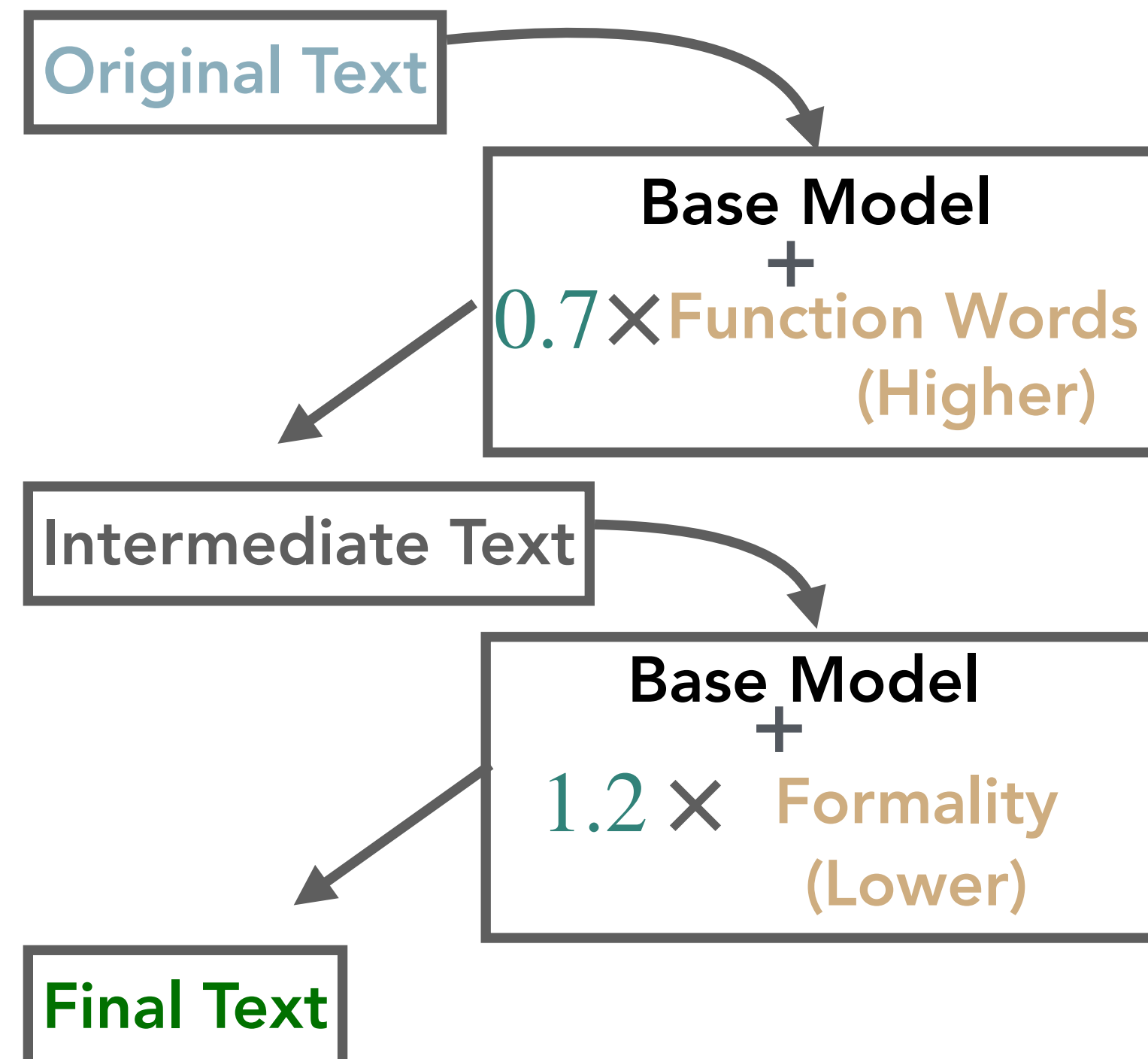
Formality
(Lower)

3. Choose Weights of
Style Axes

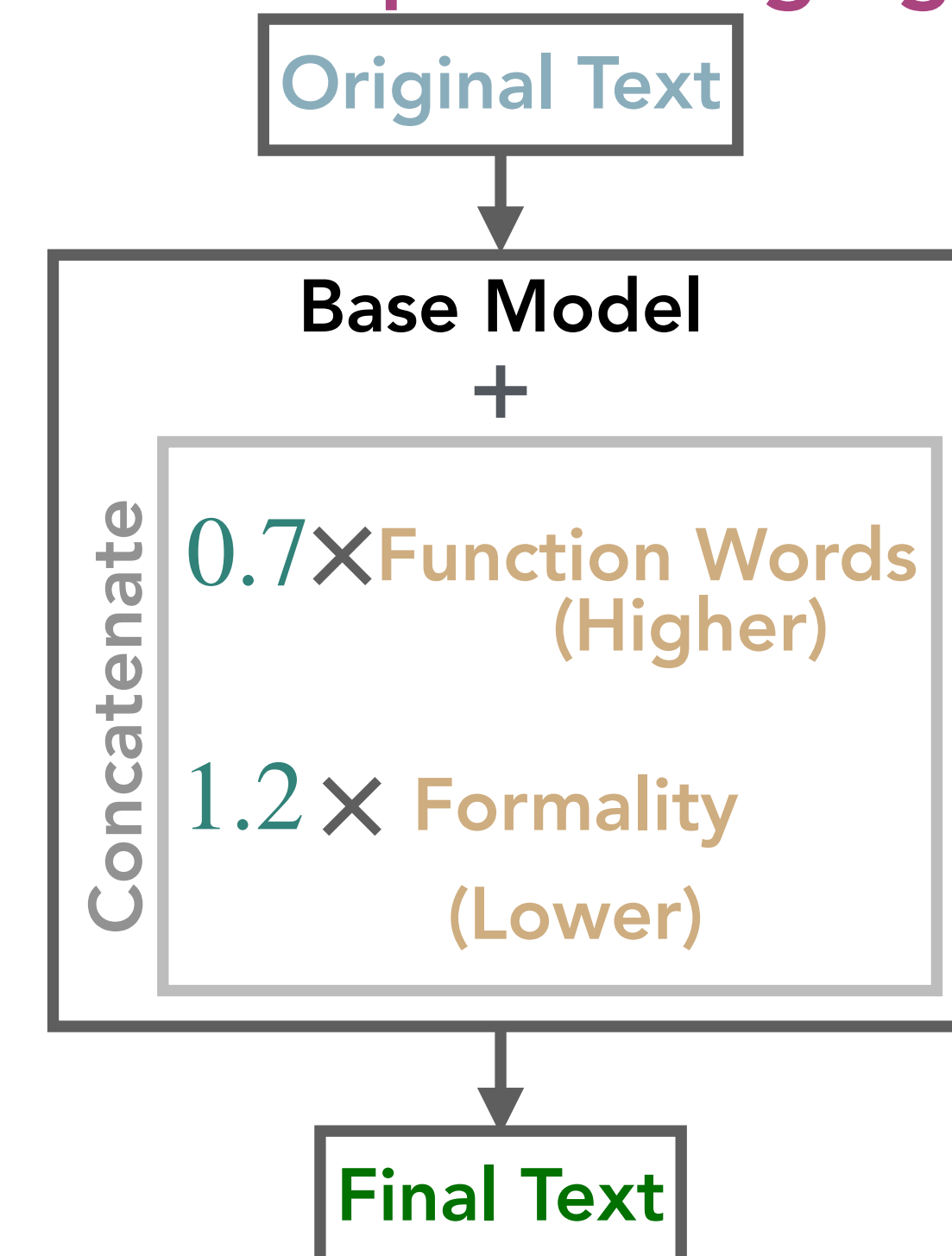
0.7, 1.2

4. Combine Style Adapters

4.a Sequential



4.a Adapter Merging



How does StyleRemix perform compared to other methods?



StyleRemix: Experimental Setup

• Four Datasets (AUTHORMIX)

1. Extended-Brennan-Greenstadt: collection of formal scholarly passages
2. Blog Authorship Corpus: diary-style entries from blog.com
3. Presidential Speeches: transcript of presidential speeches (Trump, Obama, Bush)
4. Novels: 1900s fiction writers (Fitzgerald, Woolf, Hemingway)

- Number of Authors: 3 or 5

• Baselines

- *Stylometric*: rule-based changes such as synonyms, number of words, punctuation, etc.
- *Round Trip Machine Translation*: English → German → French → English
- *Mutant-X*: Iteratively re-writes and combines randomly, uses internal classifier
- Paraphrase
- JAMDEC
- Collection of LLM's

- Base Model: Llama-3 (8B)

More
distinct
styles



StyleRemix: Evaluation Metrics



- Authorship obfuscation traditionally evaluated (automatically) on:

1. Obfuscation

How well does the rewritten text obfuscate the author style?

Metric: *Drop-Rate* using automatic authorship classifier (RoBERTa large)

2. Fluency

How understandable is the text?

Metric: *Probability of acceptable grammar* using CoLA model

3. Content Preservation

How similar in meaning is the generation to the original text?

Metric: *Cosine similarity of word embeddings*

Better for longer text

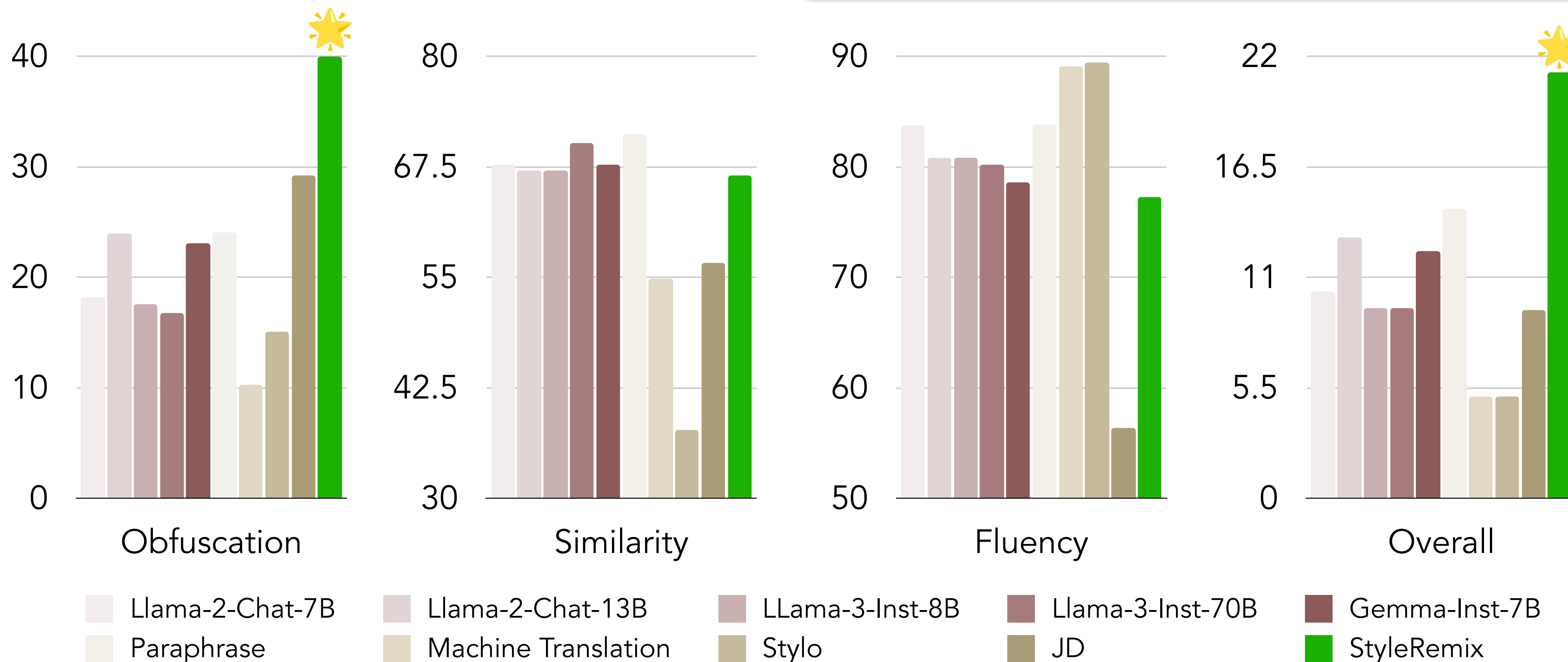
- Overall Task Score: **average** of the three metrics

$$\text{Task Score} = \frac{\text{Drop Rate} + \text{NLI} + \text{CoLA}}{3}$$

StyleRemix: Automatic Evaluation

AuthorMix - Blog

StyleRemix outperforms all baselines in obfuscation and overall quality!

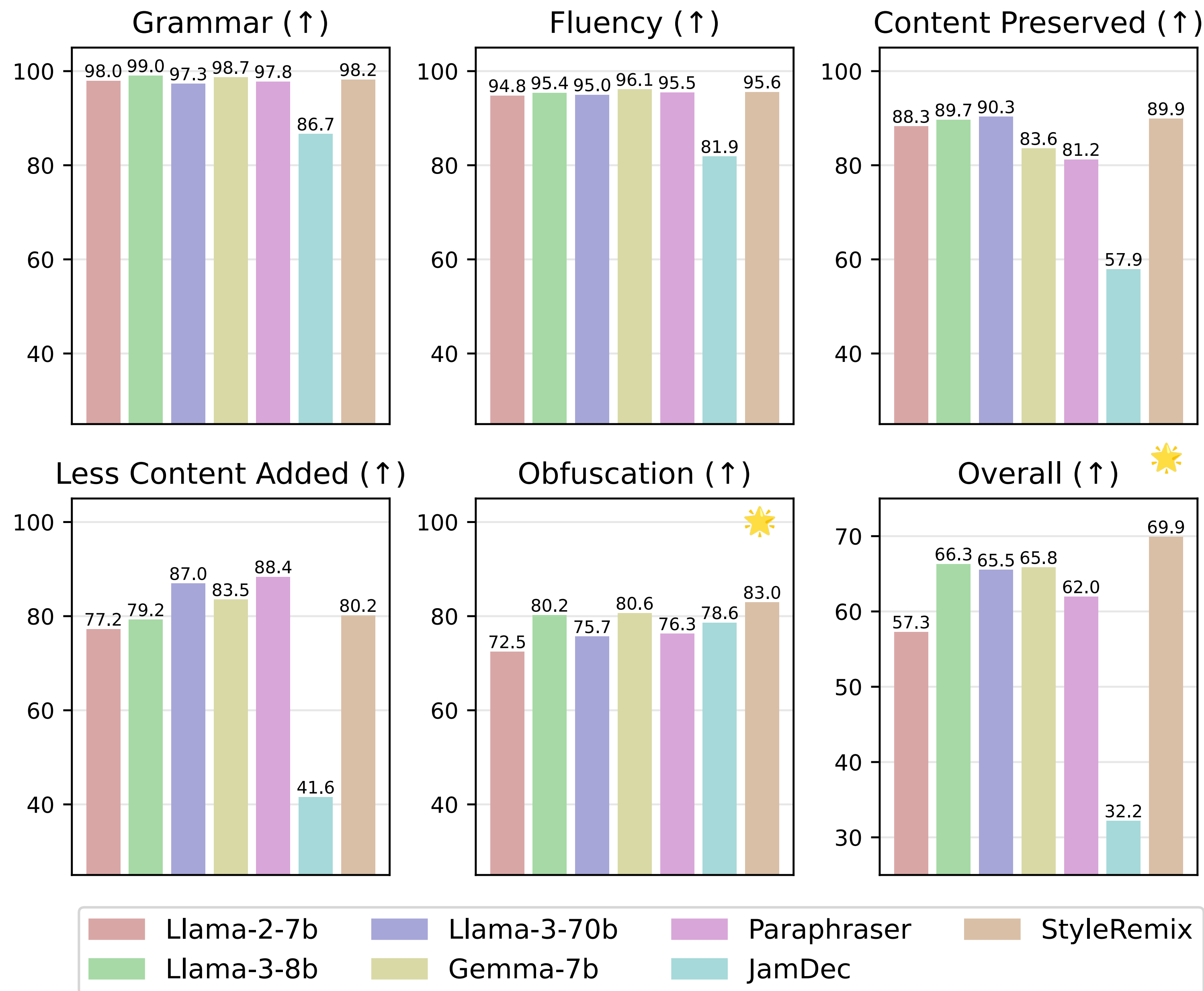


Do humans agree that StyleRemix outperforms other methods?



StyleRemix: Human Evaluation

StyleRemix has best overall obfuscation quality, even compared to much larger models!



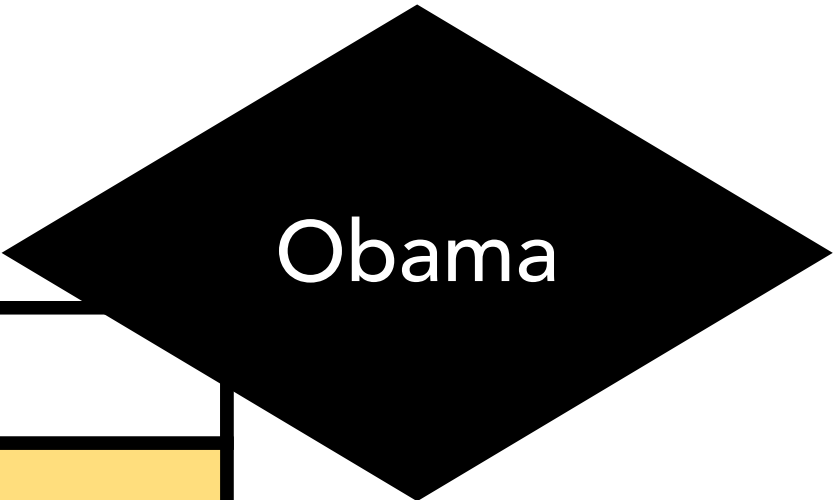
StyleRemix: Qualitative Results

Method	Generation	
Original	100%! I was surprised, but not complaining lol. But yeah Mr. Curphey's now on crutches, hobbling around the classroom and still teaching us like his usual self.	
Paraphrase	Mr. Curphey is currently using crutches in the classroom, but he is still teaching us with his usual enthusiasm and enthusiasm . I was taken aback, but I didn't feel any remorse .	Not Grammatical
Machine Translation	100 per cent! I was surprised, but I don't complain lol. But yes, Mr. Curphey is now in the gardens to walk through the classrooms and still teach us as his ordinary self.	Not Grammatical
Stylometric	One hundred percent ; i was surprised, but not complaining ll , as but yeah Mr. murphy 's now on crutches, sobbing around the classroom and always teaching us like his usual self.	Incorrect Content
JAMDEC	100%! I was surprised, but not complaining lol. But yeah Mr. Curphey's now on crutches, hobbling around the classroom and still teaching us like his usual self.	Unchanged
Llama-3 (8B)	100%! I was taken aback, but I wasn't about to grumble about it, haha! But yes, Mr. Curphey's now on crutches, limping around the classroom, and yet he's still teaching us as himself, business as usual.	Similar Style
StyleRemix	Initially, I experienced a notable degree of surprise , yet my responses remain devoid of significant complaint . Currently, Mr. Curphey, significantly, is navigating our classroom utilizing crutches , effectively maintaining his usual style of instruction, significantly.	✓ ✓ ✓

Blog

StyleRemix: Qualitative Results

Method	Generation
Original	I recognize that times have changed since I first spoke to this convention. The times have changed, and so have I. I'm no longer just a candidate. I'm the President.
Paraphrase	Since I first addressed this convention, I have seen significant changes. As a result, I have evolved from a candidate to the current President.
Machine Translation	I recognize that the times have changed since I first spoke to this convention. The times have changed, and that's how I have. I'm not just a candidate anymore. I am the President.
Stylometric	I acknowledge that times have changed since I 1st spoke to this convention, and the times have changed, and so have I. I am no longer just a candidate; and myself 'm the President.
JAMDEC	It's important to remember where we came from, where we're going, and what we're trying to do," he said, adding, "I think it's important for all of us to be. We are the people who built this country, and we have a responsibility to make sure that we preserve it for our children and our children's children, and that's what we're going to do,".
Llama-3 (8B)	Since my initial address to this esteemed convention, I have come to acknowledge that the times have indeed changed. So too have I. Gone are the days when I stood before you as a mere candidate, eager to make my case. Today, I stand before you as the President, with a wealth of experience and insight gathered during my time in office.
StyleRemix	Oh, how the world has transformed since I first addressed you all here! Indeed, the world has shifted, and so have I. It's not just about being a candidate anymore—it's about being the President.



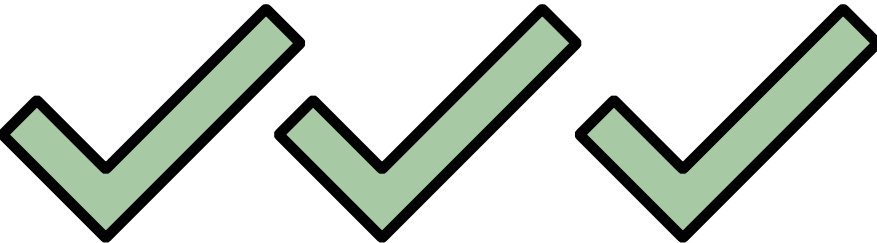
Not Grammatical

Incorrect Content

Not Grammatical

Added Information

Added Information



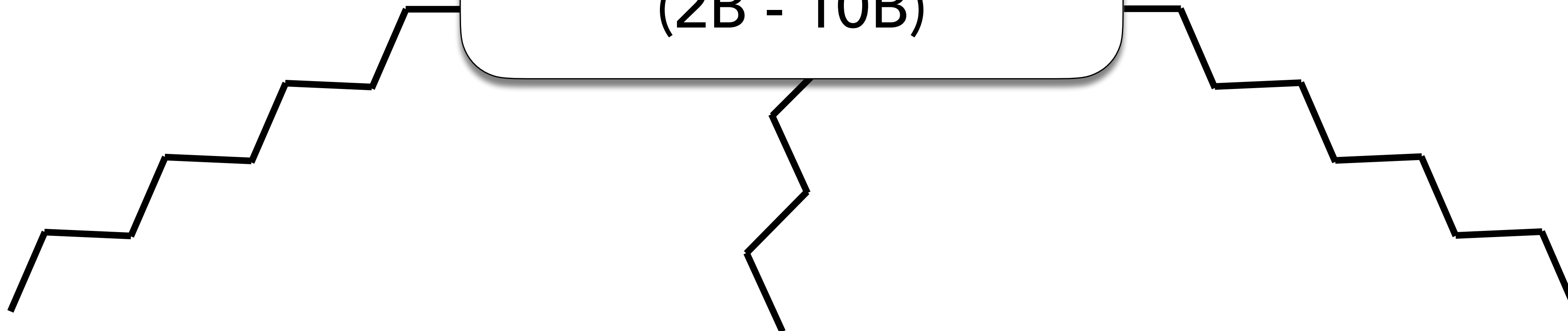
StyleRemix

Medium Models
(2B - 10B)

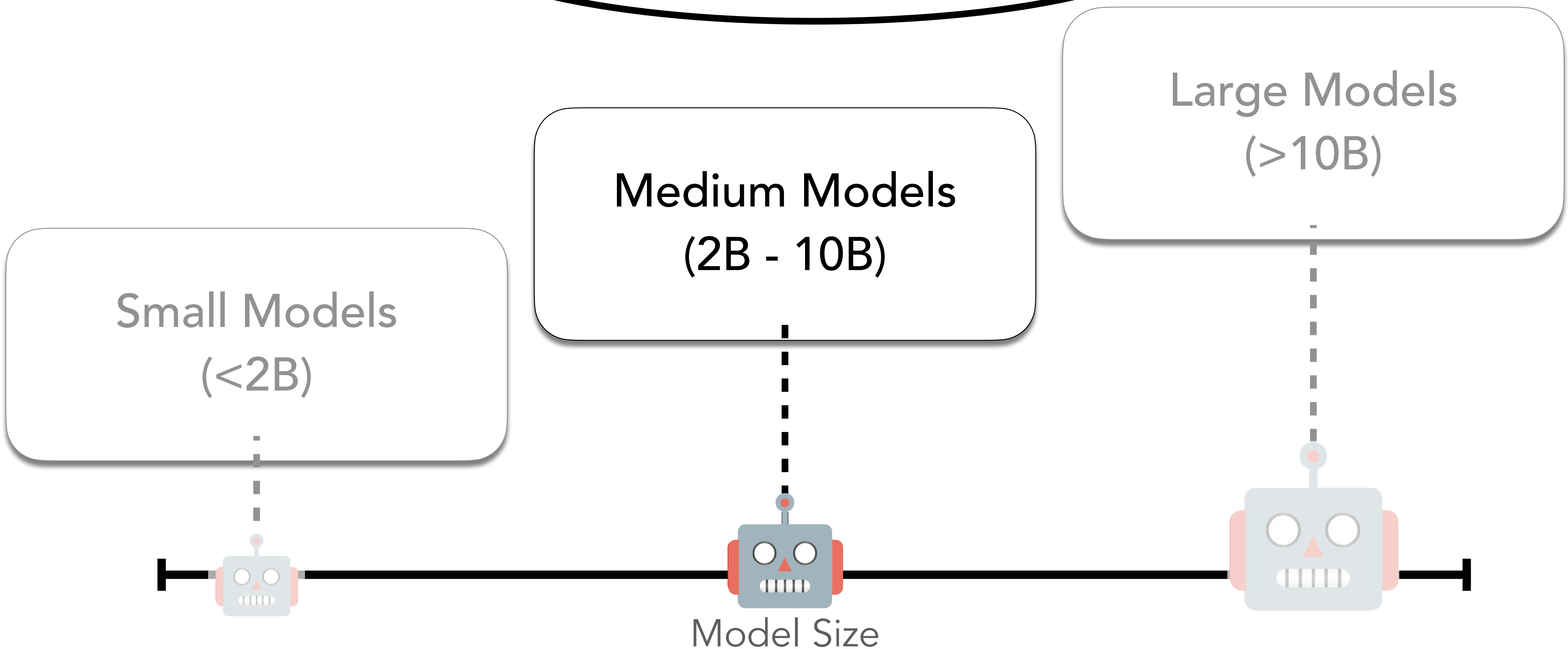
Beats Models 10X
Larger

Interpretable

Adaptive



Controllable Generation

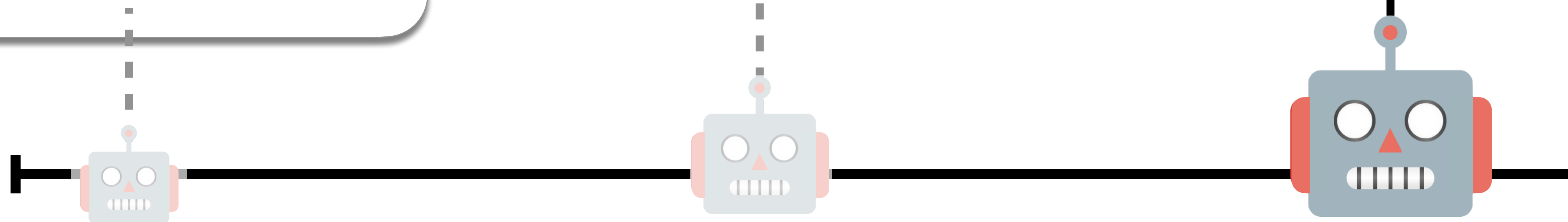


Controllable Generation

Small Models
($<2B$)

Medium Models
($2B - 10B$)

Large Models
($>10B$)



Model Size

Large Models (>10B)

Can we leverage a large language model to train a smaller model?

Pros

Good at variety of complex tasks

Higher fluency capabilities

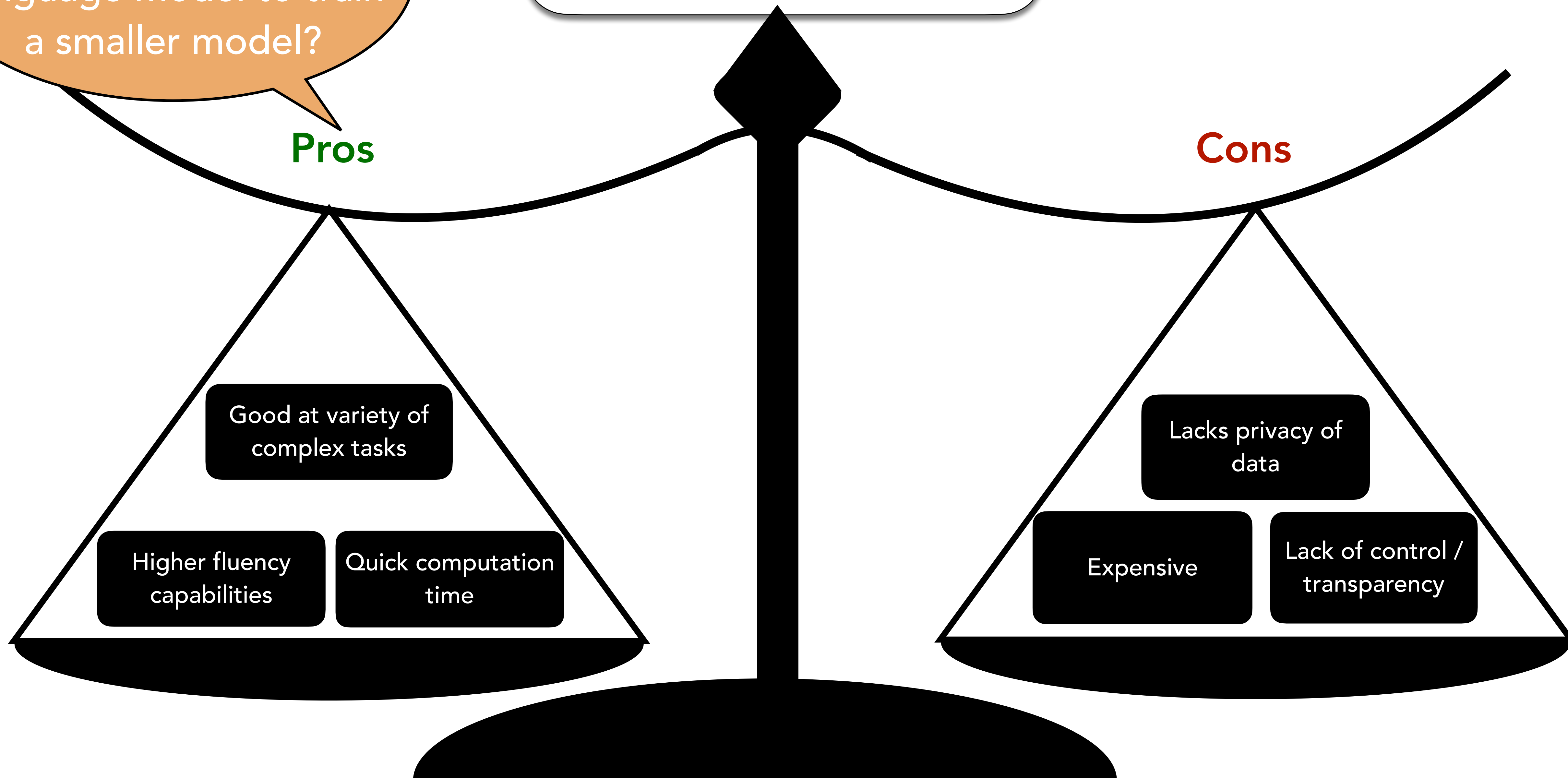
Quick computation time

Cons

Lacks privacy of data

Expensive

Lack of control / transparency



Large Models ($>10B$)

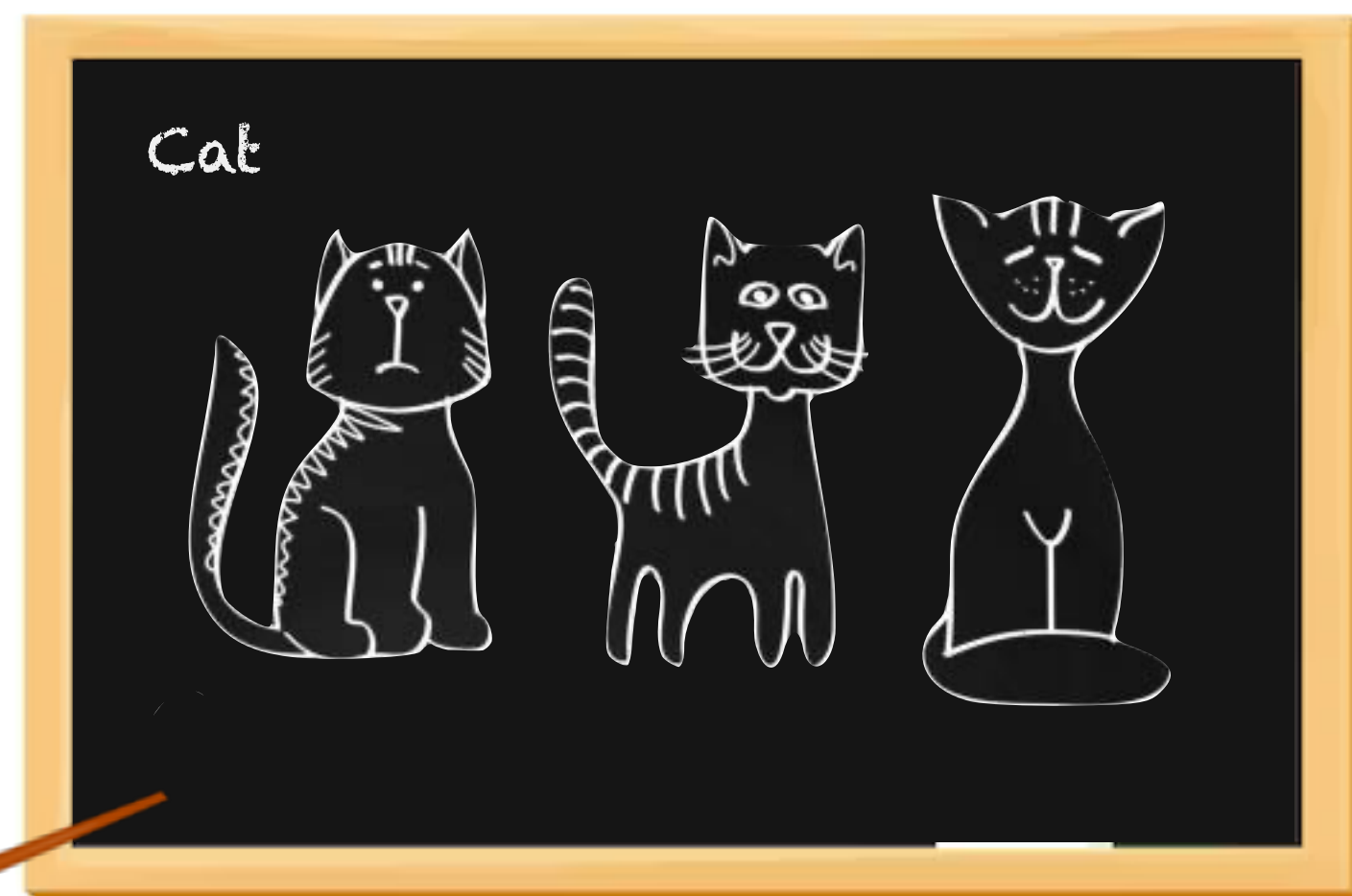
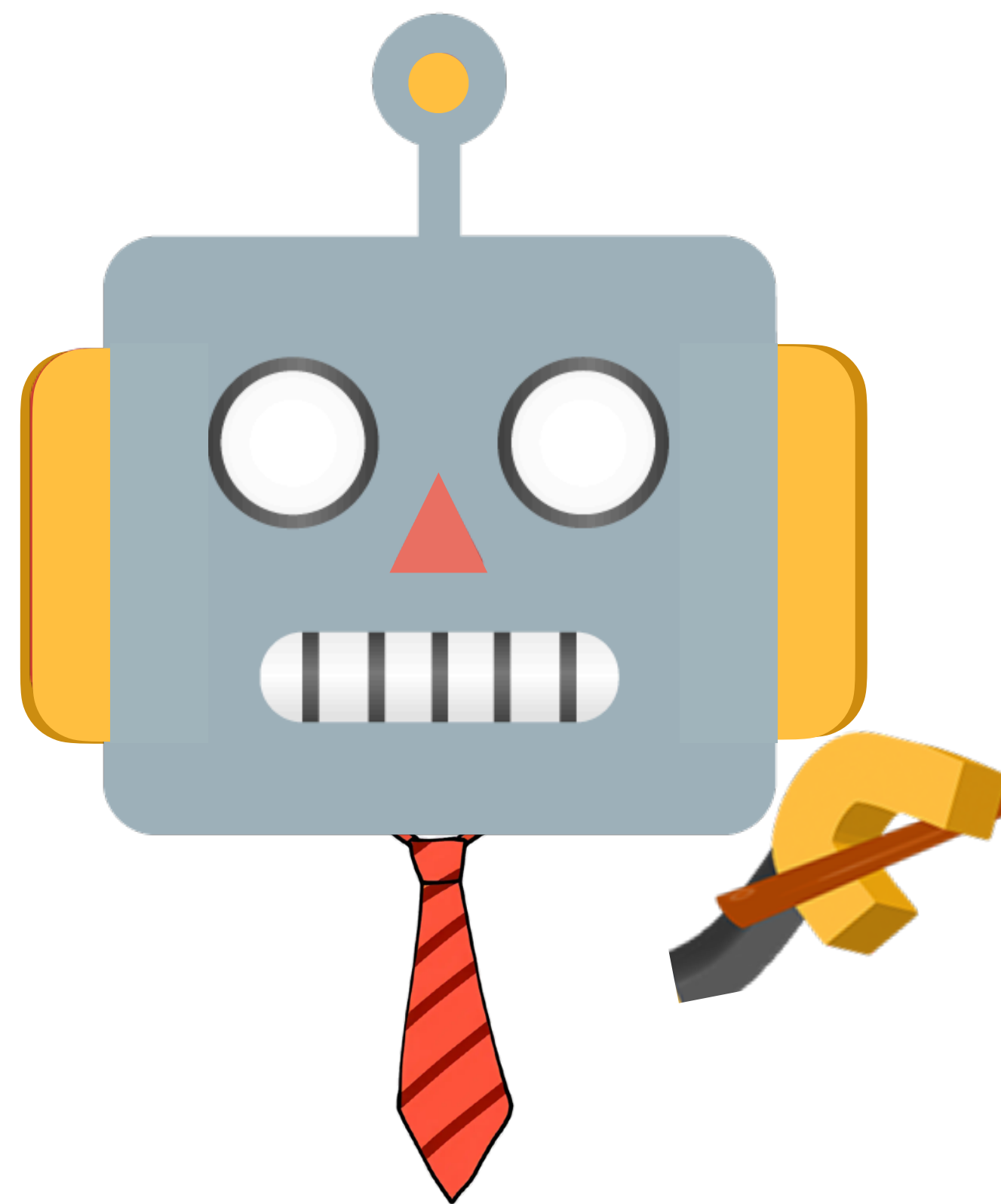
Can we
leverage a large
language model to train
a smaller model?



Yes!
Knowledge Distillation

Knowledge Distillation

Large Model
(Teacher)



Training Data

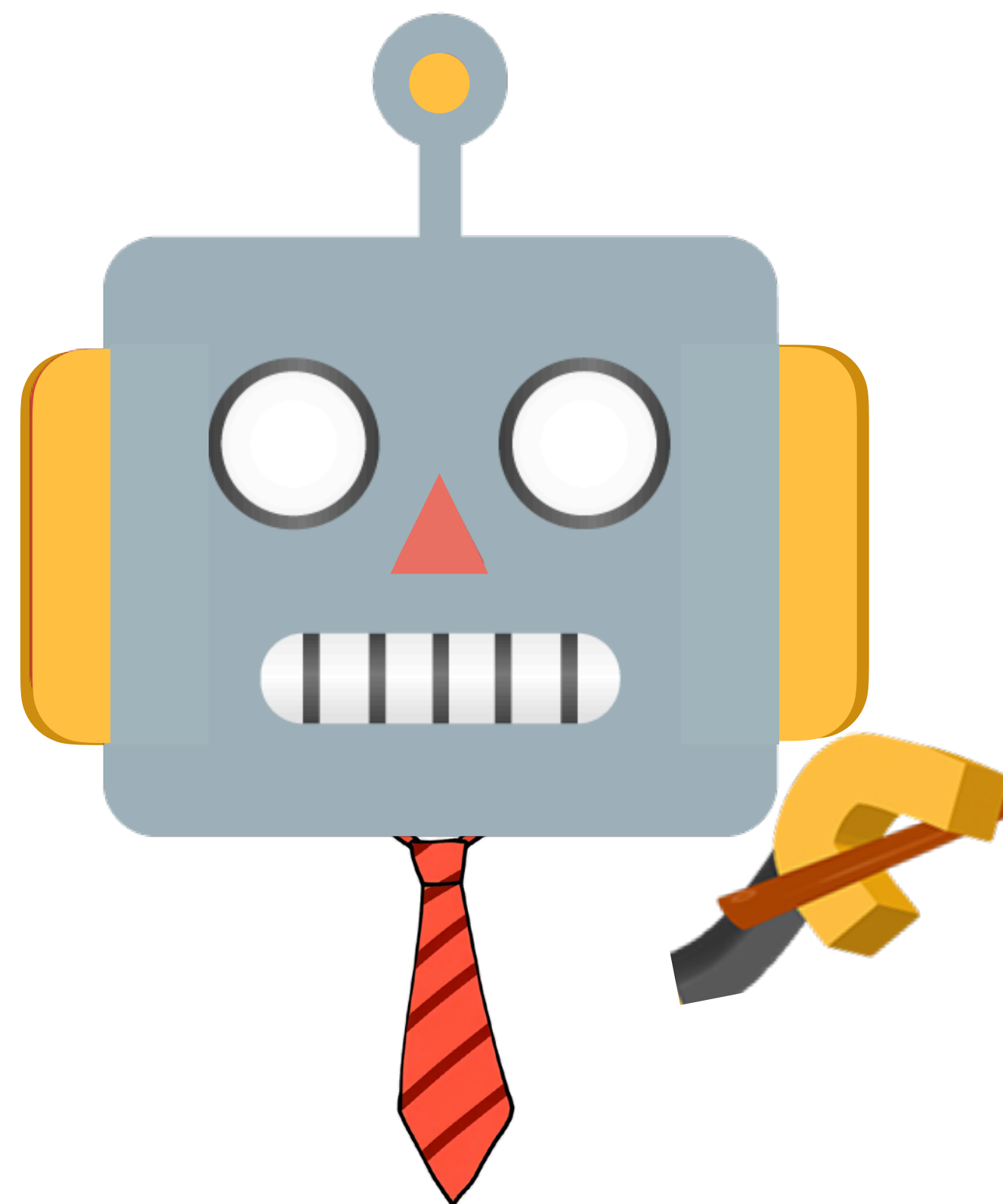


Small Model
(Student)

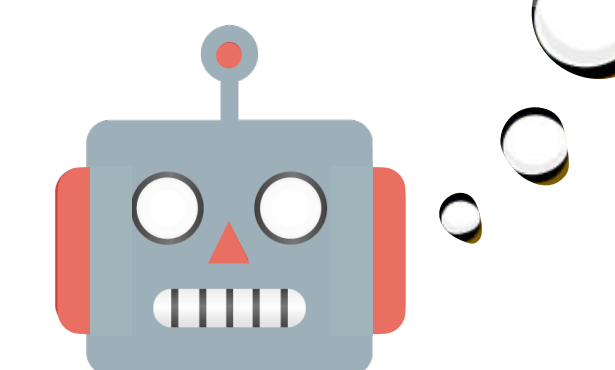
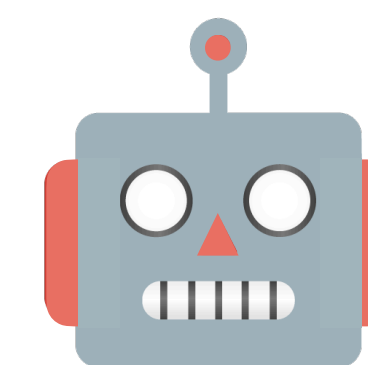
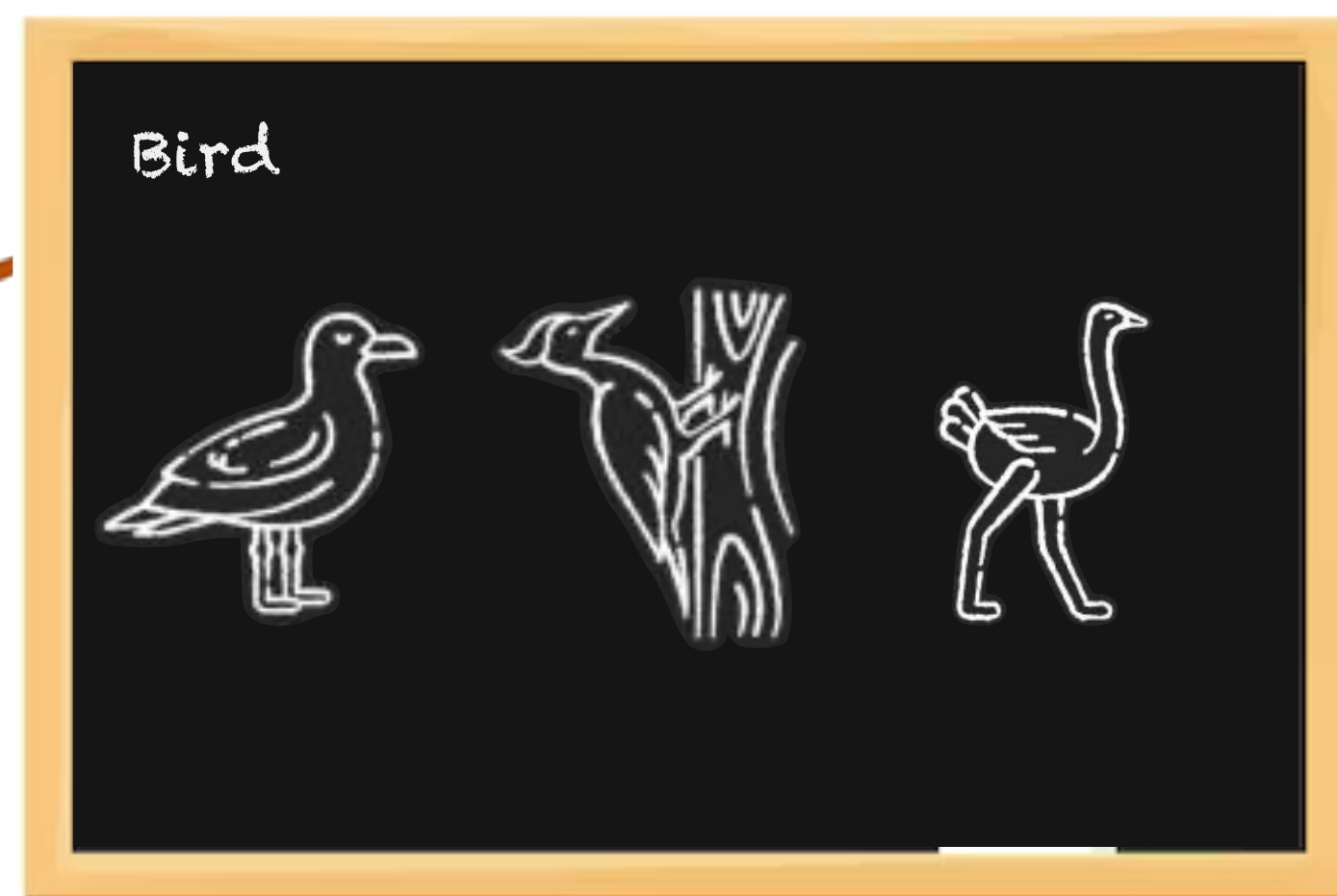
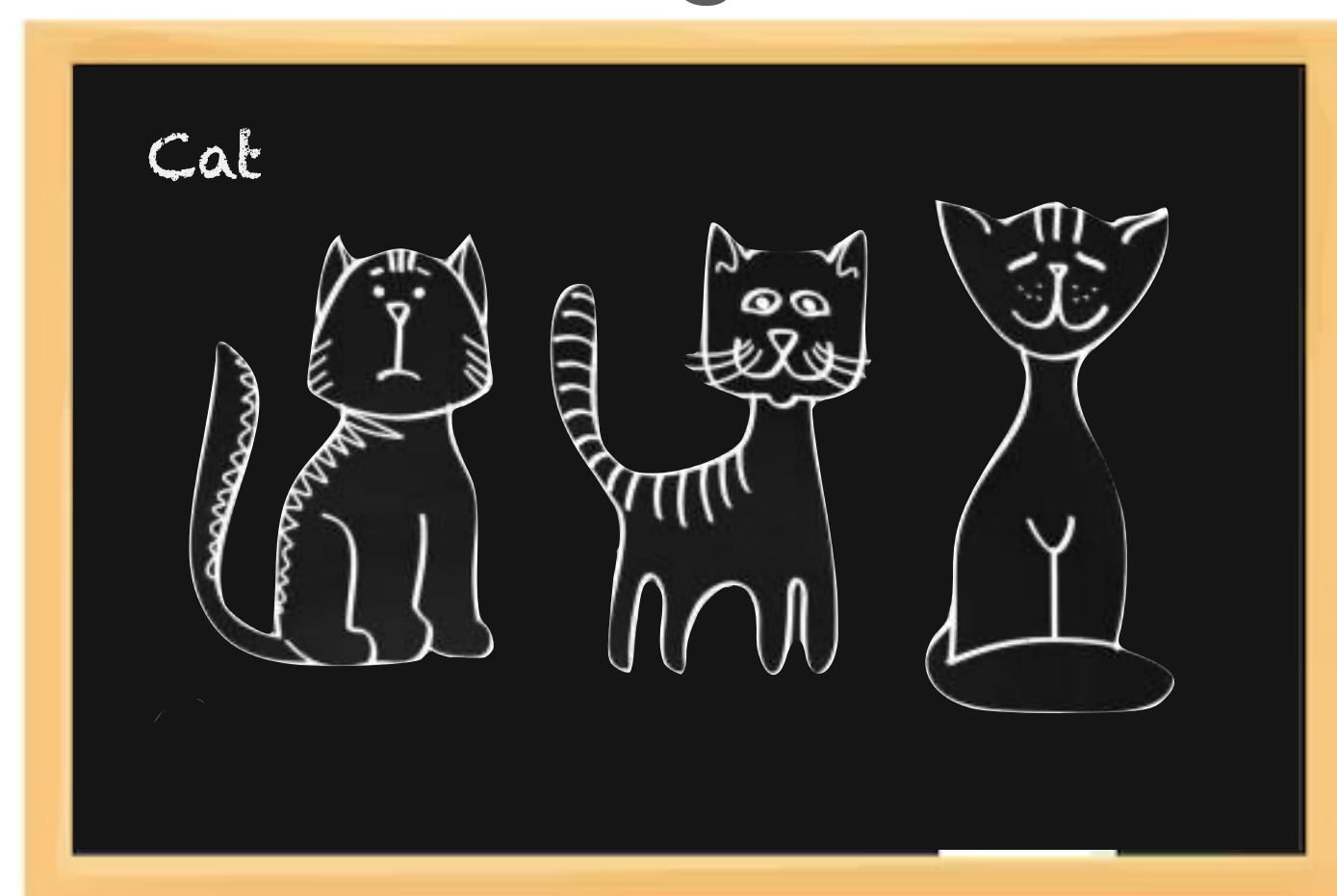
Knowledge Distillation

Double the Resources!
Doesn't utilize possible overlap!

Large Model
(Teacher)

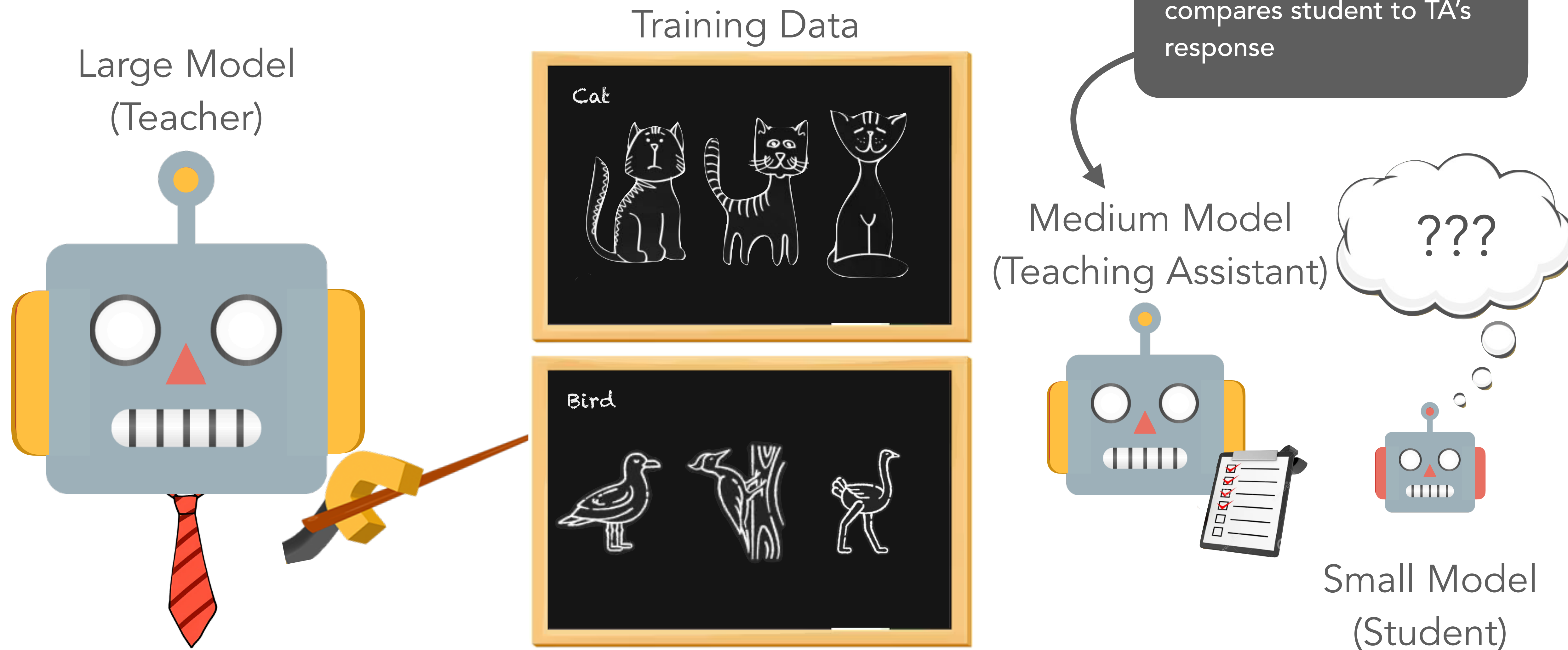


Training Data



Small Model
(Student)

Knowledge Distillation+



Knowledge Distillation+

TA Responsibilities

- Supervise that student model learns all knowledge equally
- Adaptive loss function: compares student to TA's response

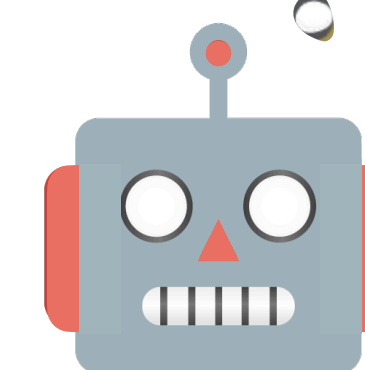
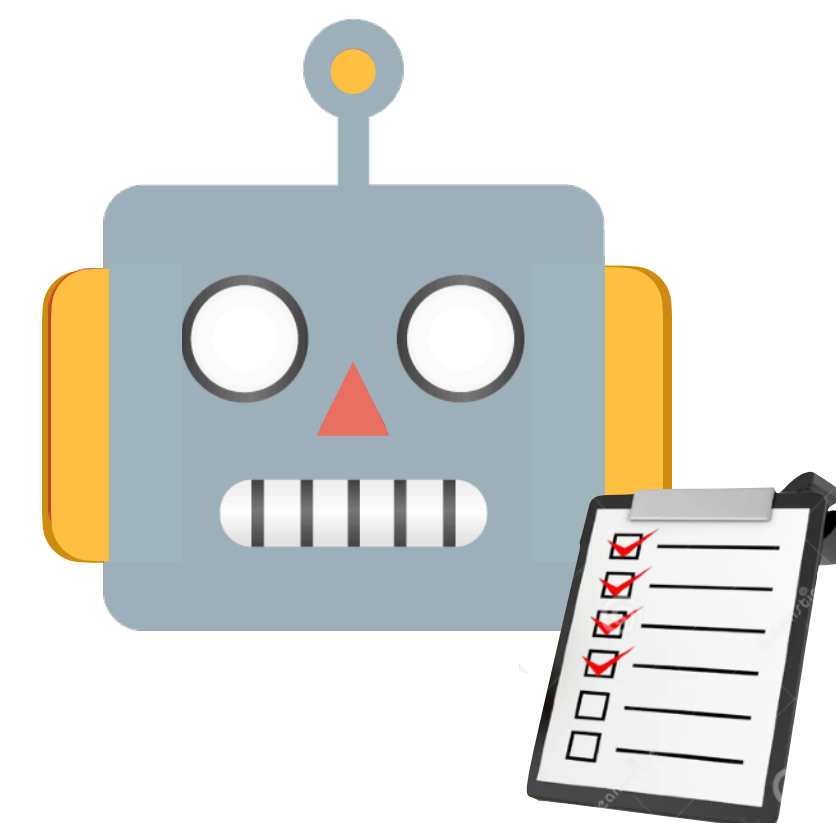
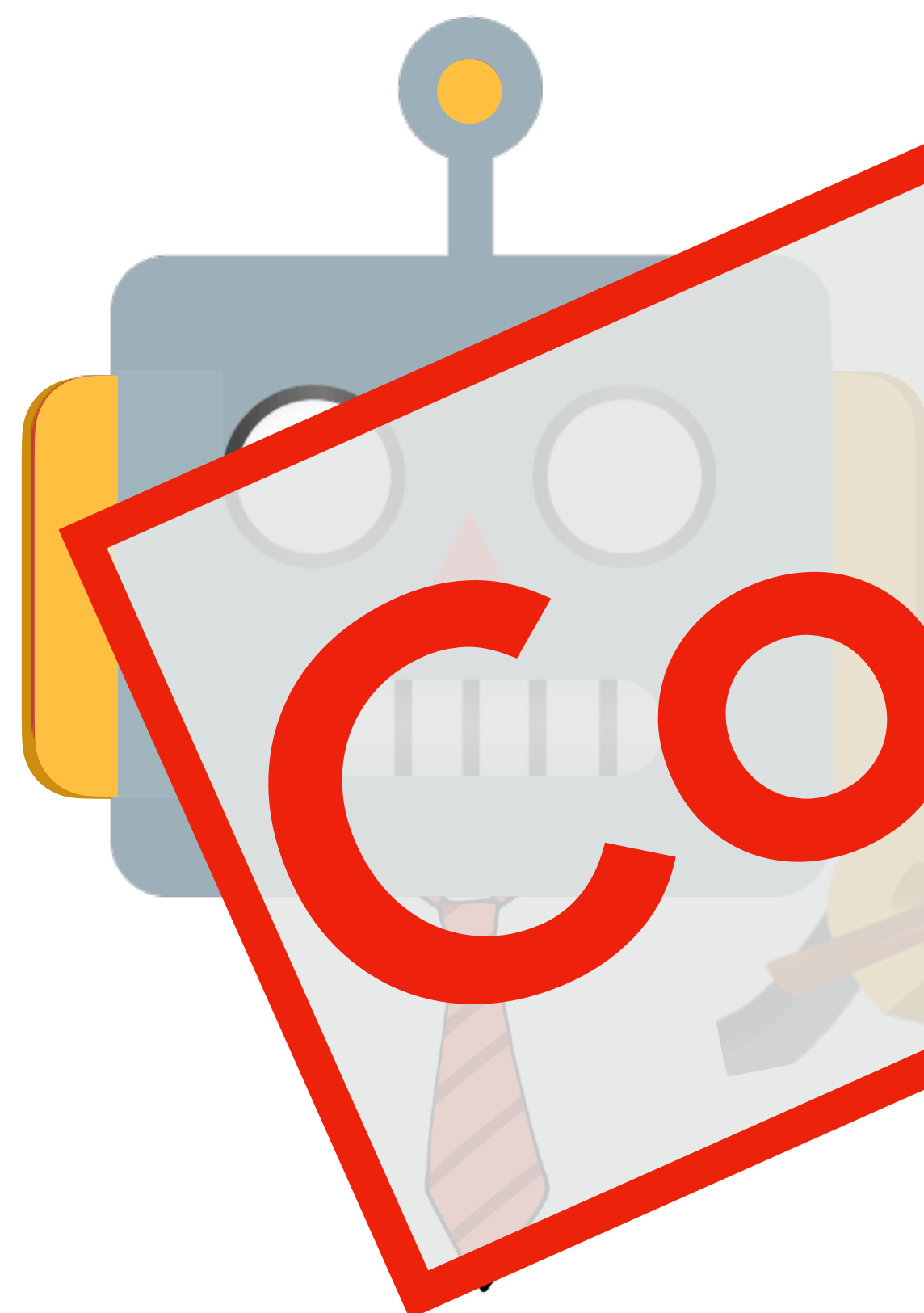
Training Data

Large Model
(Teacher)

Medium Model
(Teaching Assistant)

???

Small Model
(Student)



Thank You!

JAMDEC

Paper



Code



StyleRemix

Paper

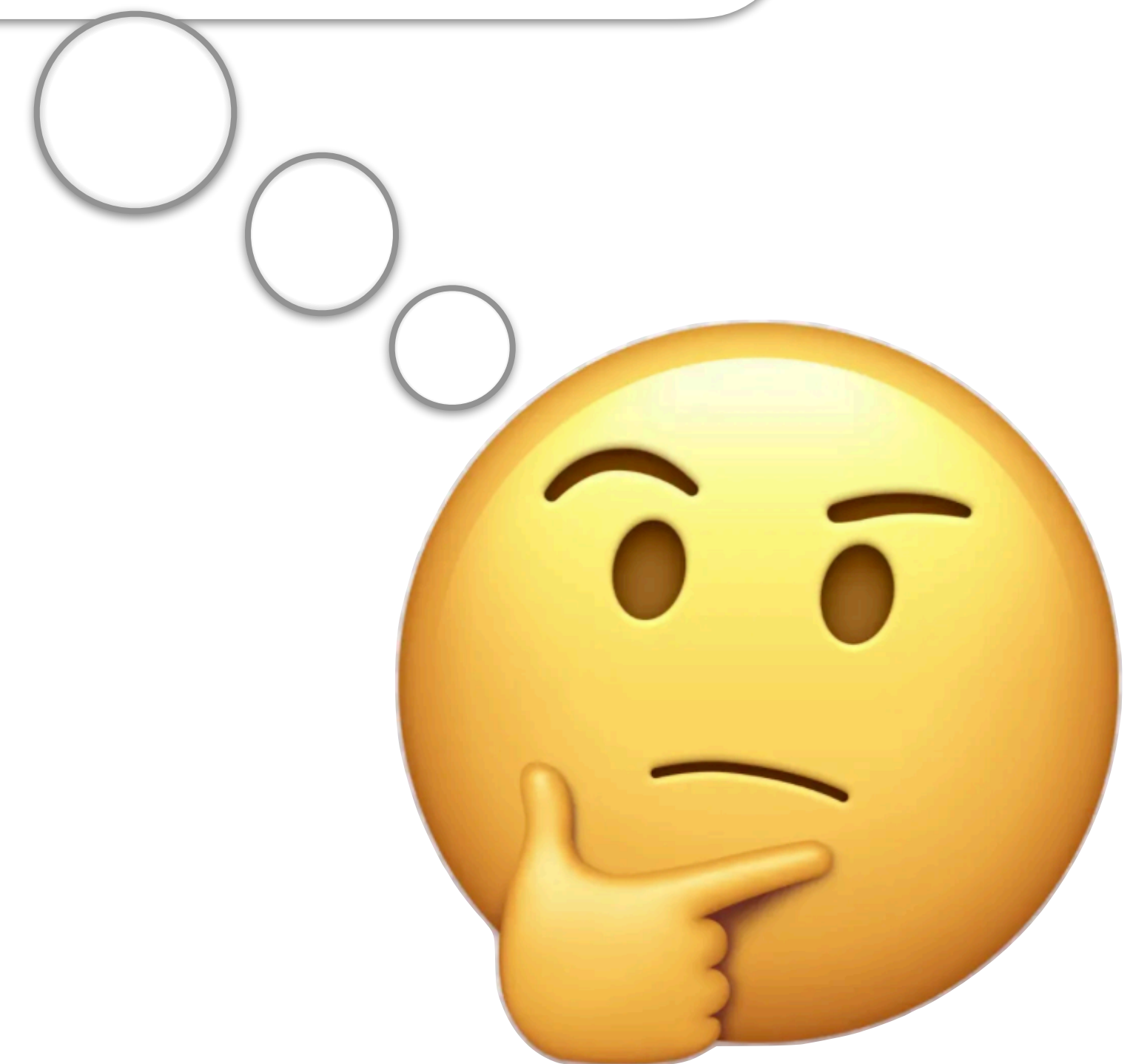


Code

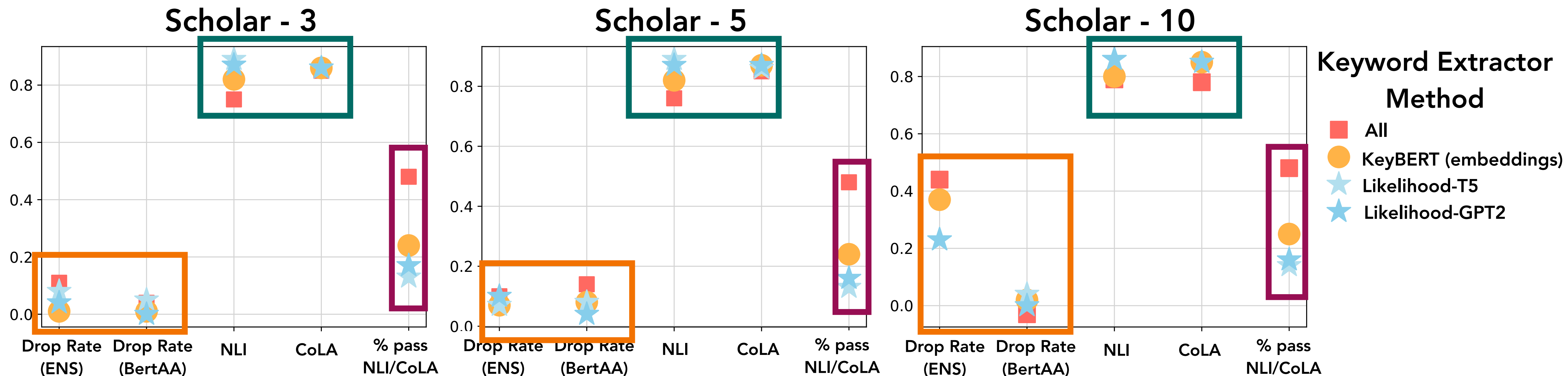


Appendix

Does our innovation to the pipeline result in better downstream performance? Likelihood Keyword Extraction? Constrained-Diversity Beam search?



JAMDEC: Keyword Extraction Comparison



All methods have similar drop rate (**Obfuscation**)

Likelihood methods have higher NLI and similar CoLA (**Fluency/Grammar**)

Using all three results in **higher % passing** NLI/CoLA threshold

↳ Each method produces diverse set of keywords

JAMDEC: Diversity Results

		JAMDEC	
Dataset	Metric	W/O Diversity	W/ Diversity
Scholar - 3	Drop Rate (ENS)	0.01	0.11
	Drop Rate (BertAA)	0.08	0.04
	NLI	0.87	0.81
	CoLA	0.86	0.79
	Average Gen.	0.16	0.52
Scholar -5	Drop Rate (ENS)	0.1	0.1
	Drop Rate (BertAA)	0.01	0.14
	NLI	0.87	0.76
	CoLA	0.87	0.85
	Average Gen.	0.16	0.48

~ 5 % increase in Obfuscation
 ~ 6 % decrease in NLI/CoLA
 ~ 35 % increases in generations
 passing NLI/CoLA threshold